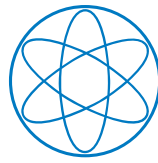




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Heralded storage of photonic polarization in a single atom

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Abstract

The long-distance communication of quantum information requires an interface that provides the ability to faithfully send and receive this type of information. In this thesis, I report on the realization of quantum-state mapping between the polarization of a photon and the spin state of a single atom, which is trapped at the centre of an optical cavity in the strong coupling regime. The implemented storage scheme is both heralded and conceptually deterministic. The storage protocol comprises three steps: First, the atomic state is prepared in a specific superposition of Zeeman levels in the ground-state manifold. Second, a faint laser pulse is reflected off the atom-cavity system which realizes a state-dependent phase shift of the combined atom-photon state. Third, detection of a reflected photon heralds a successful storage event with a demonstrated efficiency of 39 %. The inverse process, read-out of the atomic state, is realized by the reflection of a faint laser pulse and subsequent measurement of the atomic state with a measured efficiency of 35 %. The quantum mechanical character of both processes is proven by fidelities of 86 % and 88 % respectively, which is well above the classical boundary. The demonstrated novel approach to optical quantum memory opens new perspectives towards distributed quantum computation and quantum communication.

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1. Introduction

The discovery of quantum theory at the beginning of the twentieth century has revolutionized the world as we know it. The quantum theoretical framework has not only been verified by independent high precision measurements [1, 2], it has also fostered the development of devices that harness the quantum mechanical character of a given system at a macroscopic scale. Among the numerous examples of such devices are the transistor [3] and the laser [4].

Over the past decades, the discipline of quantum information processing (QIP) has emerged, which combines quantum mechanical principles with information theory in order to fundamentally enhance computability [5, 6]. Besides the efficient simulation of an inaccessible quantum system with another well-controlled quantum system [7, 8], QIP promises unbreakable encryption, a field known as quantum communication [9].

In quantum computation, the basic computational unit is a quantum-mechanical two-level system, a qubit. The possibility to entangle the states of several qubits results in a state space which scales exponentially with the number of qubits involved. It is this exponentially-scaling state space which promises a fundamental speed-up when compared to the computation with classical bits [10]. Quantum computation has garnered extensive interest after Peter Shor developed a quantum algorithm for integer factorization in a runtime that scales polynomially with increasing absolute value of the input number [11]. This algorithm could theoretically be used to break contemporary cryptosystems for communication. These systems rely on the exponential scaling of the runtime of classical factorization algorithms. Thus, a “classical” attacker can not decipher encrypted messages if the encryption key is sufficiently long [12]. The implementation of Shor’s algorithm has been demonstrated by the factorization of the number 15 in various small-scale physical systems [13, 14, 15]. However, scaling to the factorization of larger numbers remains challenging since most realizations of small-scale quantum computers rely on nearest-neighbour interactions for logic operations which makes scaling to a larger number of qubits an obstacle. A possible solution would be the use of optical photons, which can be transmitted with low loss and allow to encode quantum information in loss-independent degrees of freedom such as the polarization. They thus facilitate the distribution of quantum information over a network consisting of smaller computational nodes by quantum-state transfers [16]. In order to fulfil this task, the computational nodes within the network have to provide the ability to send and retrieve qubits encoded in photonic states [17]. Devices which provide

these qualities are referred to as optical quantum memories [18, 19].

The faithful transfer of photonic states is an important aspect in the field of quantum communication as well. The developed protocols rely on the fact that an eavesdropper listening in on the communication channel would inherently project the transmitted superposition state and therefore unveil his presence to sender and receiver [20]. While first quantum communication protocols have been demonstrated, the expansion to a global scale is hindered by the exponential loss of photons in optical fibres and thereby an exponential increase in required resources [21]. This obstacle might be overcome in a quantum repeater architecture [22, 23]. Towards this end, the communication channel is subdivided into smaller segments with nodes at each end and entanglement is distributed between adjacent nodes. Using an entanglement swapping protocol, an entangled state that is shared between sender and receiver is generated. Again, an optical quantum memory is required at each node for the intersection of the communication channel.

Due to the possible application in the aforementioned research fields, the development of an optical quantum memory has become a very active discipline. Progress has been made by storing quantum states of light in and retrieving them from atomic ensembles [24, 25, 26] and single atoms at the centre of high-finesse cavities [27, 28, 29]. The most common memory scheme, which uses atomic ensembles, is probabilistic, meaning that the success probability is inherently limited [30, 31]. However, when a successful state transfer occurred, it provides an independent measure which *heralds* a successful transfer [32, 33]. Probabilistic schemes need a longer operation time to successfully distribute entanglement in the quantum repeater scheme which is an obstacle for atomic ensembles, since inelastic scattering events might hamper the coherence time. In contrast, the approach for the state transfer onto a single atom provides long coherence times and is in principle deterministic [34]. So far, a herald for successful quantum-state transfers in a single-atom approach has not been demonstrated in practice. However, this would be highly desirable for an efficient scaling of this approach.

The goal of this thesis is therefore to demonstrate a heralded storage scheme with a single atom trapped at the centre of an optical cavity. Towards this end, a novel, deterministic atom-photon interaction mechanism [35] is employed which was recently demonstrated in our laboratory [36, 37]. It is based on the reflection of a photon off the atom-cavity system which realizes a phase shift that depends on the combined atom-photon state. Subsequent detection of the photon allows to herald the successful storage of the incident polarization state in a superposition of atomic Zeeman levels. Furthermore, the reverse process, i.e. a read-out of the atomic state onto the polarization state of a reflected photon, is demonstrated.

The results presented in chapters 3, 4 and 5.1 were obtained in close collaboration with Andreas Reiserer and are also described in his PhD thesis [38]. This thesis is organized as follows: In *Chapter 2*, the principles of optical quantum memories and procedures to

characterize their performance are introduced. The interaction mechanism as well as the schemes for storage and read-out are described in *Chapter 3*. The chapter is concluded with a quantitative treatment which concentrates on frequency detuning of the impinging photon and the resulting impact on the interaction mechanism in our setup. A prerequisite for the implementation of the scheme is full control over the internal atomic state, which is demonstrated in *Chapter 4*. The experimental results of storage and read-out of a photonic polarization qubit are described and discussed in *Chapter 5*.

2. Optical quantum memories

Single neutral atoms at the centre of high-finesse optical resonators have been identified as promising quantum memories. These systems do not only potentially exhibit long coherence times, they also allow for the individual manipulation and processing of stored information [37, 39]. The first single-atom quantum memory was recently demonstrated and fulfilled the following requirements [28]:

1. A qubit, robustly encoded in the state of an incoming light pulse, can be stored.
2. The qubit can be read-out at an arbitrary point in time.
3. The state can be read-out by mapping it onto a single photon.
4. Any qubit state can be stored and an analysis of the output pulse proves the quantum mechanical nature of the memory.

In this work, a scheme for a single-atom quantum memory is presented which has the potential to fulfil all of the requirements above. We will show the relevant steps for the storage and read-out of a photonic polarization quantum bit and prove the quantum-mechanical nature of these processes. As for the robustness of quantum states, one could think of encoding a quantum bit in the presence or absence of an excitation, e.g. the presence or absence of a photon. These implementations are unfavourable because their transfer fidelity would be significantly disturbed by unavoidable losses or imperfect detection efficiencies. We therefore employ the polarization degree of freedom which is independent with respect to loss mechanisms, as long as the beam path does not contain any polarizing elements.

Additionally, the presented scheme provides an intrinsically heralded storage procedure. This means that storage attempts yield a signal that unambiguously tells whether information was successfully stored in the atom or whether the storage attempt failed. It is important to note that this herald must not disturb the stored information, no matter which state is sent onto the memory. Detection of the herald thus circumvents two potential sources of errors. First, inevitable photon loss during transmission. Second, the inefficiencies of the storage process itself. In both cases, events in which the input photon was lost are identified.

To understand the nature of the heralding process, consider the following example for an intrinsically heralded process: spontaneous parametric down conversion [40]. Here, a pair

of photons is generated in a non-linear crystal. Detecting one photon directly heralds the presence of the second photon. Experimentalists have exploited this system to mimic an almost perfect single photon source, which lead to seminal scientific advances [41, 42].

2.1. Realization of heralded polarization storage

Before introducing the principles of our memory scheme an overview of contemporary experiments which demonstrated the heralded storage of polarization is given in this section. A detailed overview of unheralded optical quantum memories up to the year 2010 is given in [43].

The first experiment that demonstrated heralded storage of a polarization qubit was conducted in 2009 by Haruka Tanji et al. [44]. In their experiment, a cloud of atoms, which exhibit a Λ -level scheme, was trapped at the centre of an optical resonator. In order to store polarization in the ensemble, a light pulse ($\bar{n} = 500$) is applied from the side. Storage is achieved by a spontaneous Raman process which creates a collective spin-wave excitation in the cloud that corresponds to the polarization of the input pulse. This process also causes the Purcell-enhanced emission of a photon into the resonator mode. The photon is detected after it leaves the resonator and serves as a herald for a successful storage event. The efficiency of detecting a heralding photon is $\sim 10^{-6}$ per incident photon, dominated by the low free-space interaction cross section of the atomic ensemble. The stored state is retrieved by application of a resonant macroscopic laser pulse from the side and subsequent decay of the atomic excitation into the resonator mode with high efficiency (50%). The reported overlap of the input state with the read-out polarization is beyond the classical standard limit of 66%. However, one could think of a classical device which analyses the polarization of an incident light pulse with $\bar{n} = 500$ to a high fidelity. Thus, the quantum mechanical nature of this memory remains to be proven at the level of single quanta.

A second experiment which was recently published online [45] operates with a single trapped ion in order to obtain heralded polarization storage. The ion is prepared in a specific superposition state and a macroscopic beam with arbitrary polarization is applied. The spontaneous emission of the ion is collected by a lens with high numerical aperture and subsequently detected in a basis which does not infer any information of the state into which the ion decayed, thus serving as a herald for a successful storage event. The observed state fidelities are beyond the classical standard limit. Unfortunately, no input photon number is given but an input power of 12 μ W and therefore an impinging photon flux of $\sim 0.3 \cdot 10^{15} \frac{1}{s}$. Experiments in which the atomic state was read out by the production of a photon were not performed.

In summary, both experiments have demonstrated heralded polarization storage by free-space excitation of an atom or an atomic cloud in a well-prepared state. Yet, drawbacks

for both implementations have been unveiled, foremost the low heralding efficiency and the characterization with macroscopic pulses. In the light of these considerations, the experimental work, presented in this thesis, constitutes a significant contribution to the field because the employed scheme is conceptually deterministic. Moreover, the storage and the read-out process are characterized with faint laser pulses ($\bar{n} \ll 1$) which allow for an unambiguous proof of the quantum mechanical character of both processes.

2.2. Characterization of quantum processes

The work presented in this thesis concentrates on the characterization of quantum processes. These processes can be fully characterized by quantum process tomography. In order to implement this technique, one prepares a finite set of input states and reconstructs the output states after the process was performed which is described in Section 2.2.1. From these output states one infers the action of the process for an arbitrary input state as described in Section 2.2.2.

A quantum process can be written in the form of a linear completely positive map $\mathcal{E}$, which for example acts on an input density matrix ρ_{in} and returns the output density matrix ρ_{out} [5]

$$\mathcal{E}[\rho_{in}] = \rho_{out}. \quad (2.1)$$

The output density matrix ρ_{out} has to remain a physical state. Besides this limitation, $\mathcal{E}$ can represent any unitary or even non-unitary operation such as particle loss and decoherence. The results presented in this thesis analyse density matrices and examine a sub-ensemble of the experimental trials, i.e. events with a successfully detected photon, and therefore inherently normalize the reconstructed density matrices

$$\rho_{out,exp} = \frac{\rho_{out}}{\text{Tr}[\rho_{out}]}. \quad (2.2)$$

This thesis will treat density matrices of single quantum bits. We therefore simplify the following discussion to density matrices with a dimensionality of 2×2 . Considering the linearity of $\mathcal{E}$, it is convenient to express the action of $\mathcal{E}$ on single quantum bits in terms of the process matrix χ [46]

$$\mathcal{E}(\rho) = \sum_{m,n=0}^3 \chi_{mn} \sigma_m \rho \sigma_n, \quad (2.3)$$

with the Pauli operators $\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ and $\sigma_3 =$

$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. The Pauli operators $\sigma_{i=\{1,2,3\}}$ have two orthogonal eigenstates with eigenvalues ± 1 . From here on we refer to the eigenstate with positive eigenvalue as $\sigma_i |\psi\rangle = |\psi\rangle \equiv |\sigma_i\rangle$. The orthogonal eigenstate with negative eigenvalue is denoted as $\sigma_i |\psi\rangle = -|\psi\rangle \equiv -|\bar{\sigma}_i\rangle$. An ideal storage process would for example map the input state of an incoming flying qubit onto a stationary qubit and would therefore perform the identity operation with vanishing process matrix elements except $\chi_{00} = 1$. A relative phase change between the two eigenstates of σ_3 results in non-vanishing elements $\chi_{00}, \chi_{30}, \chi_{03}$ and χ_{33} [43].

2.2.1. Single-qubit state tomography

The density matrix of a two-level system can be rewritten in terms of the three-dimensional¹ Stokes vector $\vec{S}$ [47]

$$\rho = \frac{\sigma_0}{2} + \frac{1}{2} \sum_{i=1}^3 S_i \sigma_i. \quad (2.4)$$

In order to reconstruct the density matrix, it is thus sufficient to measure the Stokes vector by determining its magnitude in three linearly independent directions. However, measuring a quantum state projects the state along the axis of measurement and therefore the reconstruction of a quantum state requires multiple measurements on an ensemble of identical states. In this case one has to consider the probability to find a particle in a certain state $p_{|\psi\rangle}$ [48]

$$p_{|\psi\rangle} = \frac{N_{|\psi\rangle}}{N_{|\psi\rangle} + N_{|\psi^\perp\rangle}} = \langle \psi | \rho | \psi \rangle. \quad (2.5)$$

The direction and magnitude of the Stokes vector can be determined by evaluating the probability difference for orthogonal Pauli operator eigenstates on the corresponding coordinate axis [49]

$$\begin{aligned} S_x &= p_{|\sigma_1\rangle} - p_{|\bar{\sigma}_1\rangle} \\ S_y &= p_{|\sigma_2\rangle} - p_{|\bar{\sigma}_2\rangle} \\ S_z &= p_{|\sigma_3\rangle} - p_{|\bar{\sigma}_3\rangle}. \end{aligned} \quad (2.6)$$

An important figure of merit for the evaluation of experimental data is the fidelity of a density matrix ρ_1 with respect to another matrix ρ_2 . Initially introduced by Armin Uhlmann [50] as a transition probability, the fidelity is commonly interpreted as the overlap of two density matrices with $0 \leq F(\rho_1, \rho_2) \leq 1$ [51]

¹The Stokes parameters are generally expressed as four quantities, where the additional value corresponds to the “intensity”, in this context the number of events. Since we consider probabilities this quantity is equal to one.

$$F[\rho_1, \rho_2] = \text{Tr} \left[\sqrt{\sqrt{\rho_1} \rho_2 \sqrt{\rho_1}} \right]^2. \quad (2.7)$$

If one of the matrices is a pure state, e.g. $\rho_1 = |\psi_1\rangle\langle\psi_1|$, the fidelity reduces to the probability of finding ρ_2 in the state $|\psi_1\rangle$, see equation (2.5). The fidelity of orthogonal pure states is 0 and the fidelity of a density matrix with itself returns 1.

2.2.2. Quantum process tomography

A standard approach for the reconstruction of $\mathcal{E}$ is implemented by measuring the action of the process on four linearly independent input states, i.e. a set of vectors forming a complete basis of all 2×2 density matrices. This allows for the analytical inversion of equation (2.3) and the reconstruction of the map $\mathcal{E}$ [5]. Another approach consists of solving the obtained set of linear equations numerically with the maximum likelihood technique (see Appendix A). In an experimental setting, one is predominantly interested in the performance of the realized process $\mathcal{E}$ with respect to an ideal process $\mathcal{D}$. One way to compare both processes is to calculate the average state fidelity for all pure input states $\overline{\mathcal{F}} = \int_{|\psi\rangle} d\psi F(\mathcal{D}(|\psi\rangle\langle\psi|), \mathcal{E}(|\psi\rangle\langle\psi|))$. Assuming a trace-preserving process, the average state fidelity can be determined by using six pure input states forming a regular octahedron or four pure input states forming a regular tetrahedron in Stokes space [52]

$$\overline{\mathcal{F}} = \frac{1}{N} \sum_{\substack{\text{tetrahedron/} \\ \text{octahedron}}} F[\mathcal{D}[|\psi\rangle\langle\psi|], \mathcal{E}[|\psi\rangle\langle\psi|]]. \quad (2.8)$$

Here, N denotes the number of input states (four or six). A related measure is the process fidelity, which compares the experimentally determined process matrix χ to an expected process matrix Λ [53]

$$F_{proc}[\chi, \Lambda] = \text{Tr} \left[\sqrt{\sqrt{\chi} \Lambda \sqrt{\chi}} \right]^2. \quad (2.9)$$

The previous equation simplifies to $F_{proc} = \chi_{ii}$ if the process Λ corresponds to the action of one of the Pauli matrices σ_i . The process fidelity is related to the average state fidelity by [54]

$$\overline{\mathcal{F}} = \frac{1}{3} (2F_{proc} + 1). \quad (2.10)$$

Non trace-preserving processes

When the loss channels are independent of the input state, the experimental process $\mathcal{E}_{exp}$ can be disassembled into two subsequent processes. One process representing the normalized measurement $\mathcal{E}$, which can be extracted with equation (2.2), and an isotropic loss process $\mathcal{E}_l = p\mathbb{1}$ with $p = \text{Tr}[\mathcal{E}_{exp}]$,

$$\mathcal{E}_{exp}[\rho_{in}] = \mathcal{E}_l[\mathcal{E}[\rho_{in}]]. \quad (2.11)$$

Yet state-dependent loss channels, e.g. polarizing elements in the beam path of a photon, do not allow for such a decomposition of processes. In this case, two standard approaches exist for the characterization of the process. First, one could extract a non-uniform loss process by independently determining the particle loss probabilities for different qubit states and reconstructing a trace-preserving process after decomposition of $\mathcal{E}_{exp}$. The results obtained with this method are dependent on the measured efficiencies which might be subject to intensity fluctuations. Second, a general characterization is obtained without the knowledge of state-dependent loss channels by using six input states, which form three mutually unbiased bases, e.g. the eigenstates of the Pauli spin operators. Subsequently employing the maximum likelihood method, which approximates the normalized output states [see equation (2.2)] for ideal input states, allows for the reconstruction of a physical process [43, 54].

A non trace-preserving map reduces the trace of output density matrices and therefore the overall probability to measure the particle in an anticipated qubit state. This can be written in terms of a semidefinite positive Hermitian operator $\mathcal{P}$ [55]

$$\text{Tr}[\mathcal{E}[\rho]] = \text{Tr}\left[\sum_{n,m} \chi_{mn} \sigma_m \rho \sigma_n\right] = \text{Tr}\left[\left(\sum_{n,m} \chi_{mn} \sigma_n \sigma_m\right) \rho\right] = \text{Tr}[\mathcal{P}\rho]. \quad (2.12)$$

The 2×2 matrix operator $\mathcal{P}$ can be decomposed in the basis of its eigenstates $|\lambda_i\rangle$: $\mathcal{P} = \sum_i \lambda_i |\lambda_i\rangle \langle \lambda_i|$ and equation (2.12) becomes

$$\text{Tr}[\mathcal{P}\rho] = \lambda_1 \langle \lambda_1 | \rho | \lambda_1 \rangle + \lambda_2 \langle \lambda_2 | \rho | \lambda_2 \rangle. \quad (2.13)$$

The eigenvalues λ_1 and λ_2 therefore give the survival probability of a particle in the respective eigenstate. Using normalized output states for the reconstruction of the process matrix, intrinsically extracts an isotropic loss channel and we therefore choose to normalize one of the eigenvalues $\lambda_1 = 1$. Hence, the state $|\lambda_1\rangle$ ($|\lambda_2\rangle$) has the highest (lowest) success probability. By comparison of physically detected events and the damping prediction from the fitted process $\text{Tr}[\mathcal{P}\rho]$ for various input states, one is able to qualitatively verify the damping behaviour of the fitted process.

3. Cavity-mediated atom-photon interaction

3.1. The interaction mechanism

Interactions between atoms and photons at the level of single quanta are very weak in free space due to their low interaction cross section. The experiments in the field of cavity quantum electrodynamics (cavity QED) overcome this barrier by confining light in a small mode volume V using a high-quality optical resonator and thus increasing the interaction strength [56, 57] $g_0 = \frac{|\vec{d} \cdot \vec{E}|}{\hbar} = \sqrt{\frac{\omega \mu^2}{2\epsilon_0 V \hbar}}$ for an atomic transition with resonance frequency ω and strength μ . In this section, we present a heuristic explanation for an atom-photon interaction mechanism in a cavity QED setup, which is then used to develop a protocol for an optical quantum memory.

Consider a single-sided optical resonator, where one of the mirrors is perfectly reflecting. A single photon is impinging onto the resonator through the second mirror with nonzero transmission. At the centre of the resonator resides an atom in the strong coupling regime, i.e. the coupling strength g_0 is larger than the electric field decay rate of the resonator, κ , and larger than the atomic dipole decay rate, γ . The atom has three relevant energy levels: two stable, non-degenerate levels $|\uparrow\rangle$ and $|\downarrow\rangle$, which might be hyperfine split ground-state levels, and an excited state $|e\rangle$, see Figure 3.1 a. The atomic transitions depend on the polarization, such that only right-circular polarization (denoted as $|R\rangle$) induces transitions from the state $|\uparrow\rangle$ to the excited state, which is on resonance with the cavity frequency. In contrast, left-circularly polarized photons ($|L\rangle$) of the same frequency do not couple to the excited state of the atomic level structure.

Reflection of a resonant photon off the atom-cavity system can be fully described by considering four basic input states. If the atom is in the far detuned state $|\downarrow\rangle$, an impinging photon enters the resonator and is reflected independent of its polarization. When light is reflected off a resonant cavity it acquires a phase of π . In this case, the atom-photon product state acquires a global phase of π , which is also depicted in Figure 3.1 b. The same holds for an $|L\rangle$ -polarized input photon which does not couple to the excited state. In contrast to the previous cases, an $|R\rangle$ -polarized photon does not enter the resonator when the atom is in the resonant state $|\uparrow\rangle$, because a strongly-coupled atom-cavity system does not fulfil the resonance condition of the uncoupled system [58]. Here, the atom-photon

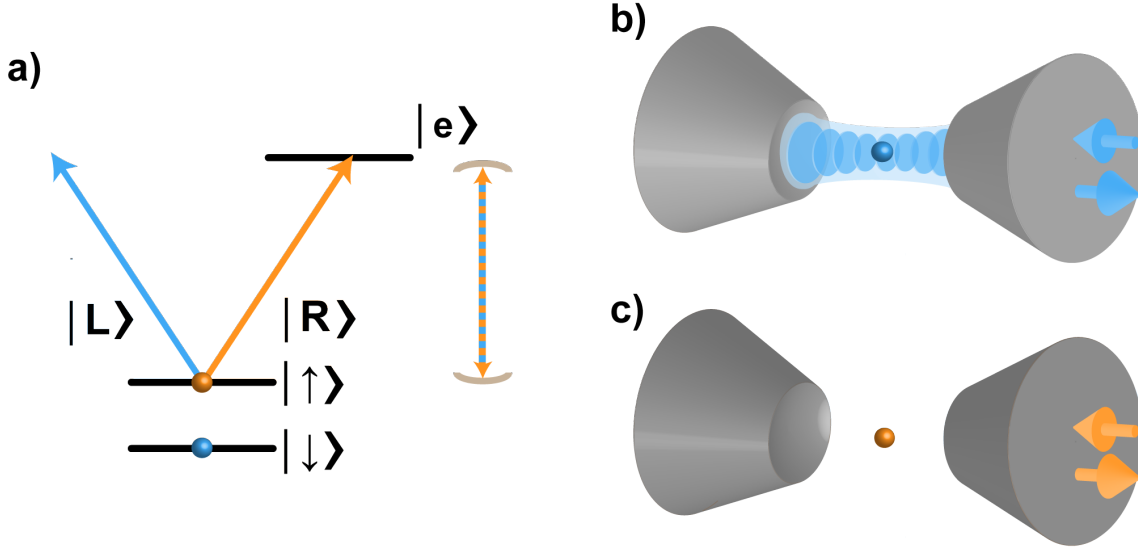


Figure 3.1.: **Illustration of the interaction mechanism.** **a)** Level scheme of the atomic state and its coupling to $|R\rangle$ -polarization on the transition $|\uparrow\rangle \rightarrow |e\rangle$. The frequency of the optical resonator is set to the transition frequency of $|\uparrow\rangle \rightarrow |e\rangle$ and the frequency of impinging photons fulfils this resonance condition as well. **b)** If the atom is in the off resonant state $|\downarrow\rangle$, photons enter and leave the resonator through the same mirror and thereby acquire a phase of π . **c)** If the atom is prepared in $|\uparrow\rangle$, an $|R\rangle$ -polarized photon is directly reflected off the atom-cavity-system and acquires no phase shift.

product state does not acquire a phase, which is illustrated in Figure 3.1 c. The action of the system is summarized in four relations with a relative phase shift of π , where $\mapsto$ symbolizes the reflection process:

$$\begin{aligned}
 |L \downarrow\rangle &\mapsto |L \downarrow\rangle \\
 |R \downarrow\rangle &\mapsto |R \downarrow\rangle \\
 |L \uparrow\rangle &\mapsto |L \uparrow\rangle \\
 |R \uparrow\rangle &\mapsto -|R \uparrow\rangle.
 \end{aligned} \tag{3.1}$$

This set of relations describes a conditional atom-photon interaction, which is generally referred to as a controlled-phase gate between two quantum bits (qubits) in the context of quantum information processing [5]. This type of gate is a universal two-qubit gate, i.e. a gate which exhibits a non-trivial interaction between two particles and, in combination with single qubit operations, allows for the realization of any quantum algorithm [59, 60].

In contrast to the original proposal [35], which used a polarizing beam splitter for the implementation of this interaction, the presented approach does not require phase-stability

of the optical setup. The interaction mechanism is coherent, meaning that superpositions are processed without projection into one of the four previously discussed cases. Expressing linear polarization in circular components leads to a balanced superposition with relative phase ϕ : $(|L\rangle + e^{i\phi}|R\rangle)/\sqrt{2}$. If the atom is in the coupled $|\uparrow\rangle$ state, any impinging linearly polarized photon will undergo a polarization flip to the orthogonal linear polarization upon reflection. As an example:

$$|D\uparrow\rangle = \frac{1}{\sqrt{2}}(|L\rangle - i|R\rangle)|\uparrow\rangle \mapsto \frac{1}{\sqrt{2}}(|L\rangle + i|R\rangle)|\uparrow\rangle = |A\uparrow\rangle. \quad (3.2)$$

If the atom is prepared in $|\downarrow\rangle$, both polarization components enter the resonator and no polarization flip occurs. This leads to the relations for the interaction mechanism in the D/A polarization basis:

$$\begin{aligned} |\downarrow D\rangle &\mapsto |\downarrow D\rangle \\ |\downarrow A\rangle &\mapsto |\downarrow A\rangle \\ |\uparrow D\rangle &\mapsto |\uparrow A\rangle \\ |\uparrow A\rangle &\mapsto |\uparrow D\rangle. \end{aligned} \quad (3.3)$$

3.2. Quantum memory protocol

In the following, a scheme for the heralded storage of polarization is presented, which builds up on the interaction mechanism of the previous Section 3.1. Suppose a photon impinges onto the atom-cavity system in an arbitrary polarization state $|\varphi\rangle_p = \alpha|D\rangle + \beta|A\rangle$, with normalized complex coefficients α and β . The storage operation is supposed to transfer the photonic state into a corresponding atomic state $|\varphi\rangle_a = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$. To this end, the atom is prepared in a balanced superposition $(|\downarrow\rangle + |\uparrow\rangle)/\sqrt{2}$ and the photon is reflected off the atom-cavity system

$$\frac{1}{\sqrt{2}}|\varphi\rangle_p(|\downarrow\rangle + |\uparrow\rangle) =$$

$$\frac{1}{\sqrt{2}}(\alpha|D\rangle + \beta|A\rangle)(|\downarrow\rangle + |\uparrow\rangle) \mapsto \frac{1}{\sqrt{2}}(\alpha|D\downarrow\rangle + \beta|A\downarrow\rangle + \alpha|A\uparrow\rangle + \beta|D\uparrow\rangle) = \quad (3.4)$$

$$= \frac{1}{\sqrt{2}}\{(\alpha|\downarrow\rangle + \beta|\uparrow\rangle)|D\rangle + (\alpha|\uparrow\rangle + \beta|\downarrow\rangle)|A\rangle\} = \frac{1}{\sqrt{2}}(|\varphi\rangle_a|D\rangle + \sigma_1|\varphi\rangle_a|A\rangle). \quad (3.5)$$

A projective measurement of the reflected photon in the D/A polarization basis projects the state to $|\varphi\rangle_a$ or $\sigma_1|\varphi\rangle_a$, depending on the measurement outcome. The action of the Pauli matrix σ_1 is a unitary rotation of the atomic state and therefore reversible. Rotating the atomic state conditioned on the detected polarization leads to a storage scheme which is heralded by the photon detection and in principle deterministic.

In addition to the storage scheme, the system also allows for the reverse process, i.e. the transfer of the atomic state onto the polarization of a photon. Suppose the atom is prepared in an arbitrary state $|\varphi\rangle_a = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$ and a $|D\rangle$ -polarized photon impinges onto the cavity. Then the atom-photon state after reflection of the photon is

$$(\alpha|\downarrow\rangle + \beta|\uparrow\rangle)|D\rangle \mapsto \alpha|D\downarrow\rangle + \beta|A\uparrow\rangle. \quad (3.6)$$

In order to transfer the photonic polarization state into the desired superposition $|\varphi\rangle_p = \alpha|D\rangle + \beta|A\rangle$, the atomic state has to be rotated by a Hadamard operation [5]:

$$\begin{aligned} |\downarrow\rangle &\mapsto \frac{1}{\sqrt{2}}(|\downarrow\rangle - |\uparrow\rangle) \\ |\uparrow\rangle &\mapsto \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle). \end{aligned} \quad (3.7)$$

The resulting atom-photon state is thus

$$\begin{aligned} &\frac{1}{\sqrt{2}}\{\alpha|D\rangle(|\downarrow\rangle - |\uparrow\rangle) + \beta|A\rangle(|\downarrow\rangle + |\uparrow\rangle)\} = \\ &= \frac{1}{\sqrt{2}}\{(-\alpha|D\rangle + \beta|A\rangle)|\uparrow\rangle + (\alpha|D\rangle + \beta|A\rangle)|\downarrow\rangle\} = \end{aligned} \quad (3.8)$$

$$= \frac{1}{\sqrt{2}}(|\varphi\rangle_p|\downarrow\rangle + \sigma_3|\varphi\rangle_p|\uparrow\rangle). \quad (3.9)$$

In analogy to the storage scheme, one can detect the atomic spin state to project the polarization of the reflected photon either into the desired state $|\varphi\rangle_a$ or a rotated version $(\sigma_3|\varphi\rangle_a)$. However, a heralding process does not exist for this read-out scheme, since the atom is assumed to be permanently trapped and could be rotated and measured at any given time even if no photon was reflected off the resonator.

At first glance, the read-out scheme seems to be a drawback to the implementation of the system as a quantum memory because it is not heralded. Consider a quantum network with nodes that exhibit this type of memory. Performing a state transfer from one node to another node is performed by reflecting a photon off the first node (read-out) and subsequently reflecting it off the second node (storage). Since the storage process is intrinsically heralded, the whole quantum-state transfer is heralded which includes failed read-outs and transmission losses. Therefore, our quantum memory is a viable option for the implementation in a quantum network, in spite of the unheralded read-out procedure.

3.3. Theoretical treatment of the reflection process

3.3.1. The empty resonator

The previous section introduced the working principles of quantum-state storage and read-out by reflection of a single photon. In the following sections, these considerations are extended to a physical system in which the second mirror is not perfectly reflecting and the resonance condition between the frequency of the impinging photon and the empty cavity might not be fulfilled. As a first step, an empty resonator is modelled using input-output theory [61]. Consider a two-sided resonator with coupling rates of the infinitely thin mirrors κ_1 and κ_2 and losses κ_l . A coherent field described by the field operator a_{in} impinges onto the left mirror (see Figure 3.2) and is partially reflected (transmitted), which is described by the field operator a_{ref} (a_{trans}). The intra-cavity field is described by the action of the harmonic annihilation operator a . In the Heisenberg picture, one formulates boundary conditions for both mirror surfaces in order to describe the system [62]

$$\begin{aligned} a_{\text{ref}} &= a_{\text{in}} - \sqrt{2\kappa_1} \cdot a \\ a_{\text{trans}} &= \sqrt{2\kappa_2} \cdot a. \end{aligned} \quad (3.10)$$

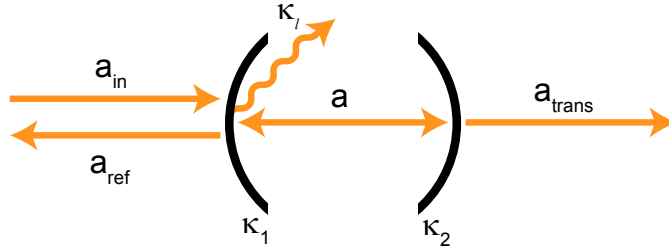


Figure 3.2.: Illustration of an optical resonator with the relevant operators and mirror parameters for a treatment using input-output theory.

In a Heisenberg-Langevin treatment of the system, the equation of motion is

$$\frac{\partial a}{\partial t} = -\frac{i}{\hbar} [a, \mathcal{H}_{\text{sys}}] - (\kappa_1 + \kappa_2 + \kappa_l)a + \sqrt{2\kappa_1}a_{\text{in}}. \quad (3.11)$$

Here, the Hamiltonian $\mathcal{H}_{\text{sys}} = \hbar\omega_c a^\dagger a$ describes an empty cavity with resonance frequency ω_c . Fourier transformation of equations 3.10 and 3.11, allows for the calculation of the reflected and transmitted intensities in dependence of the impinging field frequency ω

$$|a_{\text{ref}}|^2[\omega] = \left(1 - \frac{4\kappa_1(\kappa_2 + \kappa_l)}{(\omega - \omega_c)^2 + \kappa^2}\right) |a_{\text{in}}|^2 = R_{\text{id}}[\omega] |a_{\text{in}}|^2 \quad (3.12)$$

$$|a_{\text{trans}}|^2[\omega] = \left(\frac{4\kappa_1\kappa_2}{(\omega - \omega_c)^2 + \kappa^2} \right) |a_{\text{in}}|^2 = T_{\text{id}}[\omega] |a_{\text{in}}|^2. \quad (3.13)$$

Reflected and transmitted intensities are thus Lorentzian functions with a full width at half maximum of $2(\kappa_1 + \kappa_2 + \kappa_l) = 2\kappa$. Measuring transmission and reflection of the optical resonator in our laboratory therefore allows for the reconstruction of the mirror parameters. However, in an experimental situation the geometric overlap of an impinging light pulse and the resonator mode is not perfect. The unmatched fraction of light is reflected by the first mirror, which has a transmission coefficient on the order of ppm [54]. Therefore, it is a good approximation to assume that this reflection is perfect and the altered reflection coefficient of the system becomes $R[\omega] = \eta + (1 - \eta) \cdot R_{\text{id}}[\omega]$. To quantify the coupling efficiency of photons coming out of an optical fibre to the resonator mode, light is transmitted through the resonator and sent onto the fibre. Estimating the coupling efficiency of light into the fibre relates to the directly reflected fraction of light and is determined to be $\eta = 8(3)\%$. The same principle applies for the transmission, which is detected in a fibre coupled detection setup. In a simultaneous measurement of reflection and transmission the detection efficiency needs to be taken into account $T[\omega] = \frac{\eta_{\text{trans}}}{(1-\eta)} \cdot T_{\text{id}}[\omega]$, with a measured coupling efficiency of $\eta_{\text{trans}} = 80(3)\%$ in our non-optimized setup.

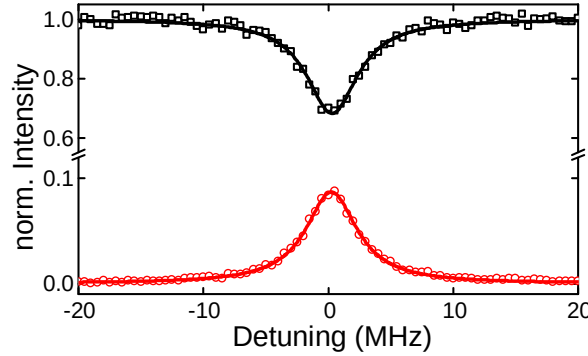


Figure 3.3.: Measured resonator transmission (red) and reflection (black). The fitted curves are derived from input-output theory. First, the reflection is fitted with the free parameters κ_1 (amplitude of the dip) and κ (linewidth). With these parameters the transmission curve is fitted by varying κ_2 . Both curves have a slight frequency offset of 0.3 MHz.

Figure 3.3 presents the measured transmission (red) and reflection (black) as a function of the frequency detuning $\omega - \omega_c$. The theoretical curves include the aforementioned coupling efficiencies and correspond to a parameter set of $(\kappa, \kappa_1, \kappa_2) = 2\pi \cdot (2.5(1), 2.3(1), 0.08(1))$ MHz. With the mirror distance of 486 μm one calculates a round-trip loss of 102(3) ppm, in good agreement with a previous ring-down measurement that yielded a loss of 103(2) ppm [54]. The transmission of the outcoupling mirror is determined to be 92(3) ppm, which devi-

ates from a transmission measurement before the resonator was assembled that resulted in 99 ppm [54]. The transmission of the high reflecting mirror is determined to be 3(1) ppm. From mirror transmissions and round-trip losses one infers the total intrinsic absorption losses to be 7(3) ppm.

One can independently verify the ratio κ_2/κ_1 by trapping atoms at the centre of the cavity and applying a resonant laser beam from the side. The atomic fluorescence is guided into the resonator mode by the Purcell effect. Monitoring the output signal behind both mirrors results in a ratio of 4.0 %. The predicted ratio from the fit above is 3.6 %, in reasonable agreement with the independently measured ratio.

Besides determining the mirror parameters, the data gives the reflection on resonance: 70(2) %. This is an important figure of merit for the success probability of the reflection based atom-photon mechanism of Section 3.1 and has to be taken into account for $|L\rangle$ -polarized photons and all cases in which the atomic state is $|\downarrow\rangle$.

3.3.2. Coupled atom-cavity system

An extension of the empty resonator can be obtained by incorporating the Jaynes-Cummings Hamiltonian into the Hamiltonian of the system with the atomic transition frequency ω_a and the coupling strength g [57]

$$\mathcal{H}_{sys} = \hbar \left\{ \omega_c a^\dagger a + \omega_a |e\rangle \langle e| + g(|e\rangle \langle \uparrow| a + |\uparrow\rangle \langle e| a^\dagger) \right\}. \quad (3.14)$$

Calculating the evolution of the system for this Hamiltonian with a Heisenberg-Langevin treatment results in a set of coupled differential equations for expectation values of the following observables: the intra-cavity field a , the polarization of the atomic state $|\uparrow\rangle \langle e|$, and the atomic state population $|e\rangle \langle e|$ [63, 64]. Assuming a weak drive, i.e. $\langle |e\rangle \langle e| \rangle = 0$, one can solve this set of equations analytically in the steady-state situation [65, 66], which results in the following equations for the field reflection $r(\omega)$ and transmission $t(\omega)$ of the system

$$r[\omega] = 1 - \frac{2\kappa_1 (\gamma + i(\omega - \omega_a))}{g^2 + (\kappa + i(\omega - \omega_c)) (\gamma + i(\omega - \omega_a))} \quad (3.15)$$

$$t[\omega] = \frac{2\sqrt{\kappa_1 \kappa_2} (\gamma + i(\omega - \omega_a))}{g^2 + (\kappa + i(\omega - \omega_c)) (\gamma + i(\omega - \omega_a))}. \quad (3.16)$$

Here, $\gamma = 2\pi \cdot 3 \text{ MHz}$ denotes the free-space atomic polarization decay rate of ^{87}Rb . These equations allow for an estimation of the measured intensity in reflection $|r[\omega]|^2$ and the phase of the field with $\text{Arg}[r[\omega]]$ [67].

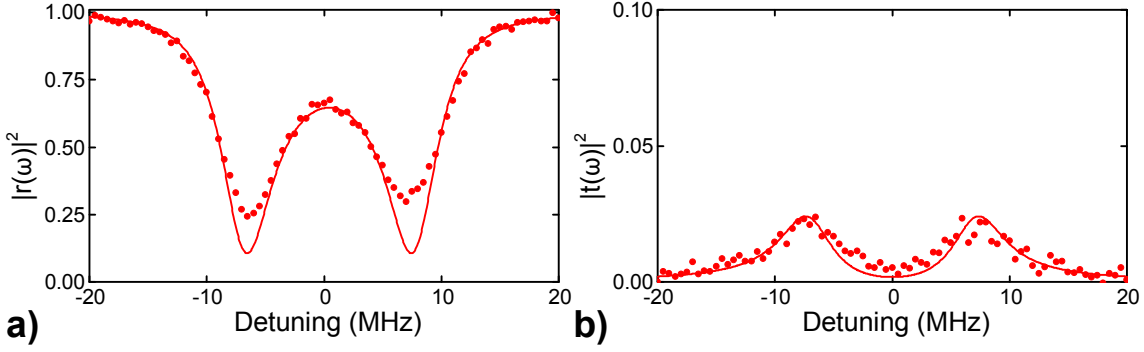


Figure 3.4.: Spectrum in the coupled case. The data points denote the reflected [a)] normalized intensity and the simultaneously recorded transmission [b)] of the resonator.

The reflected and transmitted intensity for a strongly coupled atom in our system is presented in Figure 3.4a and b respectively (red data points). The theoretical curves are derived from equations 3.15 and 3.16 with $\omega_c = \omega_a$, a coupling strength of $g = 2\pi \cdot 7$ MHz, the cavity parameters and the fibre coupling efficiencies of Section 3.3. These curves are in good qualitative agreement with the data. The main deviation on the normal-mode peaks in reflection might result from two experimental imperfections. First, fluctuations in the atomic frequency due to varying AC stark shifts might broaden and flatten these dips [38]. Second, for this measurement the atom has to be prepared in the coupled state, which is not always the case (see Section 4.2). From the measured spectrum in reflection one infers that an impinging resonant photon is reflected with a probability of 66(2) %, which is also the probability for the successful implementation of the interaction mechanism in the coupled case (see Section 3.1).

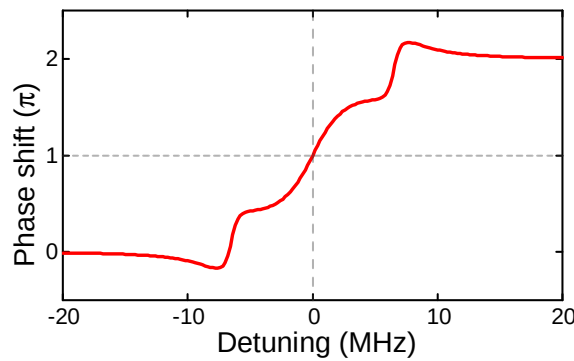


Figure 3.5.: Phase difference (red) of the reflected field between coupled and uncoupled case. The grey dashed lines indicate the desired phase difference of π on resonance.

Besides predicting reflected and transmitted intensities, equation (3.15) allows for the calculation of the phase shift between the coupled and uncoupled case ($g = 0$). Figure 3.5 shows the expected phase shift $\text{Arg}[r[\omega, g = 2\pi \cdot 7 \text{ MHz}]] - \text{Arg}[r[\omega, g = 0]]$. On resonance, a phase difference of π between coupled and uncoupled case is expected. Yet a small frequency detuning might result in varying phase differences and it seems to be important to operate exactly on resonance. This fact is experimentally investigated in Section 5.1.

3.3.3. Simple model for cavity-birefringence and its impact on the memory scheme

Resonator birefringence might have an impact on the polarization dependent atom-photon interaction mechanism. This effect might be caused by slightly elliptical mirrors, which result in different resonance frequencies. The resonator in our setup possesses an effective frequency difference of 420 kHz between the linearly polarized eigenmodes, which are close to $|D\rangle$ and $|A\rangle$ [43]. Since this difference is small compared to the resonator linewidth, one can assume that the empty resonator imprints a phase difference ξ on both eigenmodes and neglect amplitude modifications in a good approximation. In the following, $|L\rangle$ -polarized light impinges onto the resonator and it is assumed that the phase shift between both polarizations is imprinted onto the diagonal polarization component

$$|L\rangle = \frac{1}{\sqrt{2}}(|D\rangle + |A\rangle) \mapsto \frac{1}{\sqrt{2}}(e^{i\xi}|D\rangle + |A\rangle). \quad (3.17)$$

In the L/R polarization basis, this state is expressed as a rotation towards right-circular polarization

$$|L\rangle \mapsto \frac{1}{2}(1 + e^{i\xi})|L\rangle + \frac{1}{2}(1 - e^{i\xi})|R\rangle. \quad (3.18)$$

The system is further approximated by the assumption that birefringence is not a gradual process but rather rotates incident polarization as soon as light leaves the resonator. We return to the atom-photon interaction mechanism and analyse the reflection of a $|D\rangle$ -polarized photon if the atom is in the strongly coupled state, as discussed in Section 3.1. The $|R\rangle$ -polarized fraction of the impinging photon is directly reflected (see Figure 3.6), whereas the $|L\rangle$ -polarized component enters the resonator, undergoes birefringence and is rotated into an elliptical polarization state

$$\frac{1}{\sqrt{2}}(-i|R\rangle + |L\rangle)|\uparrow\rangle \mapsto \left\{ i\frac{|R\rangle}{\sqrt{2}} + \frac{1}{\sqrt{8}}\left((1 + e^{i\xi})|L\rangle + (1 - e^{i\xi})|R\rangle\right) \right\}|\uparrow\rangle. \quad (3.19)$$

The absolute phase difference in reflection for light which is resonant with the cavity frequency and light with a frequency offset of 420 kHz is computed with input-output theory [see equation (3.15)]. One finds an absolute phase difference of $|\xi| = 0.113\pi$ for our cavity

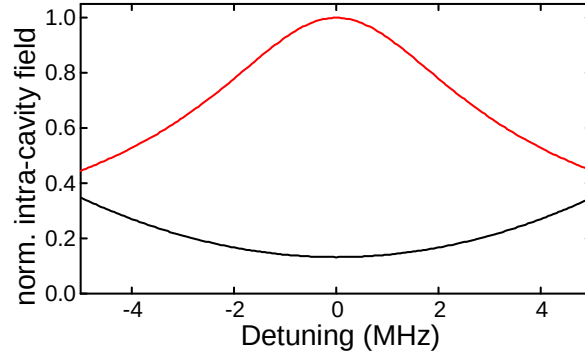


Figure 3.6.: Absolute value of the intra-cavity field $|\langle a \rangle|$ as a function of frequency detuning for the coupled (black) and uncoupled (red) case. Both curves are normalized to the maximum value of the uncoupled case and derived from the resonator parameters of Section 3.3. If an atom couples strongly to the cavity, the build up of an intra-cavity field is suppressed by an order of magnitude.

parameters. Hence, one expects to find the reflected polarization state in the coupled case to be very close to $|A\rangle$, with a predicted fidelity [see equation (2.7)] of 97 %. The asymmetry of the coupled reflection process thus results in a suppression of birefringence. This is the most simple approximation to the process. In a more tedious calculation it might turn out that the presence of a strongly coupled atom actively suppresses the rotation of $|L\rangle$ towards $|R\rangle$ by lowering the intra-cavity field amplitude of coupled light. In a simple picture, the atom scatters $|R\rangle$ -polarized light out of the resonator. A thorough calculation of this effect would need a time-resolved numerical calculation, which goes beyond the scope of this thesis.

Considering the previous estimation, the gate equations 3.1 in the D/A basis become in a good approximation

$$\begin{aligned}
 |\downarrow D\rangle &\rightarrow e^{i\xi} |\downarrow D\rangle \\
 |\downarrow A\rangle &\rightarrow |\downarrow A\rangle \\
 |\uparrow D\rangle &\rightarrow |\uparrow A\rangle \\
 |\uparrow A\rangle &\rightarrow |\uparrow D\rangle.
 \end{aligned} \tag{3.20}$$

The impact of these equations on the storage scheme [see equation (3.9)] is expressed in the expected final states with a rotation denoted as $R_z(-\xi)$ upon detection of the herald in $|D\rangle$

$$\frac{1}{\sqrt{2}}(\alpha e^{i\xi} |\downarrow\rangle + \beta |\uparrow\rangle) |D\rangle + \sigma_1 |\varphi\rangle_a |A\rangle = \frac{1}{\sqrt{2}}(R_z(-\xi) |\varphi\rangle_a |D\rangle + \sigma_1 |\varphi\rangle_a |A\rangle). \tag{3.21}$$

For the read-out scheme the rotation due to cavity birefringence manifests itself in a general rotation of the read-out polarization state along the z-axis in Stokes space

$$\frac{1}{\sqrt{2}}(R_z(-\xi) |\varphi\rangle_p |\downarrow\rangle + R_z(-\xi) \sigma_3 |\varphi\rangle_p |\uparrow\rangle). \quad (3.22)$$

We note that this rotation could be mitigated by either shifting the cavity frequency such that $|D\rangle$ -polarized photons are on resonance or by a redefinition of the quantum bits in the storage scheme and subsequent read-out with $|A\rangle$ -polarized photons.

4. Controlling and detecting the spin of a single atom

The aforementioned atom-photon interaction mechanism requires full control over the atom at the centre of the cavity. This control has been achieved in our experimental setup by implementing a three-dimensional optical lattice [68], and will be the topic of the thesis of Andreas Reiserer [38]. Since the general setup configuration [43, 54] and fundamental techniques, such as loading, trapping and cooling of a single atom [69], have been described in previous doctoral theses, the scope of this chapter is the characterization of experimental techniques required for the storage and read-out protocol of Section 3.2.

The trapping configuration and the level structure exhibited by ^{87}Rb atoms in this specific configuration are presented in Section 4.1. Initial preparation of the atom into an electronically well-defined state and its read-out are described in Section 4.2. Another essential ingredient in the implementation of the mechanism is full control over a qubit subspace of the ^{87}Rb ground-state manifold, which is demonstrated in Section 4.3.

4.1. Level structure of ^{87}Rb in a dipole trap

Neutral atoms can be trapped in optical dipole traps, which have been the subject of reviews and books (e.g. [70, 71]). Our system employs three standing-wave far-off-resonance traps (FORT), which are described in the following.

One of the beams is a 771 nm blue-detuned intra-cavity trap, which forms a standing wave-pattern along the cavity axis. The wavelength of this beam is chosen such that the intra-cavity standing-wave pattern exhibits a node at the centre of the resonator, while the standing-wave of light resonant with the D_2 absorption line of ^{87}Rb ($5S_{1/2} \rightarrow 5P_{3/2}$) has an anti-node at this position. Hence, the atom is trapped at the point of maximum coupling strength along the cavity axis. The second FORT is also a 771 nm beam, which is aligned along the vertical direction and retro-reflected after passing the resonator. The last trapping beam is a 1064 nm retro-reflected beam, which propagates horizontally and passes the resonator perpendicular to its axis. These trapping beams confine the atom in three dimensions with trap frequencies of about 400 kHz on all axes [68]. The light fields are orthogonally polarized, such that they do not interfere. Each lattice beam generates

the following interaction potential for a two level atom with transition frequency ω_{eg} and decay rate $\Gamma_{e \rightarrow g}$ [71]:

$$U_{dip}[\vec{r}] = \pm \frac{3\pi c^2 \Gamma_{e \rightarrow g}}{2\omega_{eg}^3} \left(\frac{1}{\omega_{eg} - \omega} + \frac{1}{\omega + \omega_{eg}} \right) I[\vec{r}] \quad (4.1)$$

Here, c is the speed of light, ω the frequency of the light field and $I(\vec{r})$ the intensity distribution of the beam. The interaction potential has a negative (positive) sign if the atom is in its ground-state (excited state). Thus, the action of the potential on an atom in the ground-state is attractive (repulsive) if the light field is red (blue) detuned and exactly the opposite if the atom is in the excited state. In our trapping geometry, the atomic transitions are mainly shifted by the red-detuned lattice beam, as the atom is trapped at nodes of the two blue detuned lattices and the induced level shift by these beams is therefore negligible at low temperatures.

The atom-photon interaction mechanism employed in this work requires a specific level structure (see Section 3.1), in which the atom couples to one polarization type (e.g. right-circular polarization). In order to show that ^{87}Rb exhibits this type of level structure in our trapping geometry, the energy shift for a state $|F, m_F\rangle$ has to be calculated by summing over all transitions to other states [72]. The 1064 nm lattice beam is linearly polarized along the resonator axis and the quantization axis is defined to be parallel to the polarization axis of the lattice, thus prohibiting transitions with $\Delta m_F \neq 0$.

Considering all available transitions, including their coupling strengths, with a principal quantum number of $n \leq 9$ [73, 74, 75], leads to the normalized Zeeman level shifts shown in Figure 4.1. As expected for a red-detuned trap, all Zeeman levels of the ground-state (excited state) are shifted towards lower (higher) energies. However, the AC Stark shift is non-uniform for the Zeeman levels of the $F' = 1$ and the $F' = 3$ manifold. These are the transitions which exhibit the required non-uniform substructure. For the experiments in this thesis, we choose the transition of $|2, 2\rangle \leftrightarrow |3', 3\rangle$. The coupling strength of this transition is the largest of all possible $5S_{1/2} \leftrightarrow 5P_J$ transitions (with $J = \{\frac{1}{2}, \frac{3}{2}\}$), leading to the largest coupling constant in our system [70].

One can infer the transition frequency shift of $|2, 2\rangle \leftrightarrow |3', 3\rangle$ by measuring a normal-mode spectrum (see Figure 3.4). In order to obtain such a spectrum the frequency of the cavity and the atomic transition frequency have to be on resonance. Therefore the central frequency of the measured normal-mode spectrum equals the atomic transition frequency. We find a transition frequency shift of 100 MHz with respect to the unperturbed transition. Thus, the transition for left-circularly polarized light, $|2, 2\rangle \leftrightarrow |3', 1\rangle$, is detuned by 95 MHz. This allows to implement the interaction mechanism and we define, in reference to Section 3.1, the coupled state as $|2, 2\rangle \equiv |\uparrow\rangle$. The uncoupled state $|\downarrow\rangle$ is chosen to be $|1, 1\rangle \equiv |\downarrow\rangle$, as $|1, 1\rangle$ can be coupled to $|2, 2\rangle$ using Raman beams (see Section 4.3).

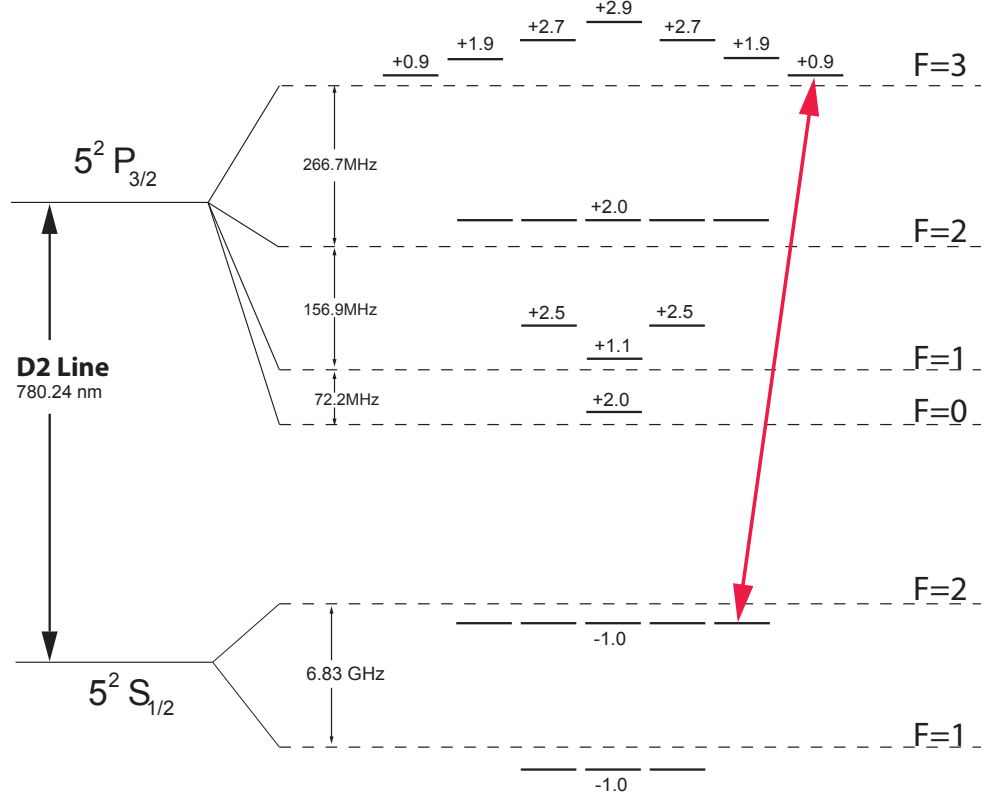


Figure 4.1.: **Predicted level structure of an ^{87}Rb atom in a π -polarized 1064 nm FORT.** The dashed lines correspond to the undisturbed level structure. The black numerical values denote the AC Stark shift of the Zeeman levels relative to the absolute AC Stark shift of the $5^2S_{1/2}$ ground-state levels. The derived values are in excellent agreement with a recent publication [76]. We identify the transition of $|2, 2\rangle \rightarrow |3', 3\rangle$ (indicated by the red arrow) as the desired transition for the implementation of the interaction mechanism.

4.2. Optical pumping and detection of the atomic state

After defining the atomic quantum bit in the level structure of ^{87}Rb , we turn to the initialization of the atom in one of the two chosen Zeeman states. To this end, optical pumping with two laser beams is employed. One beam impinges from the side and is resonant with the $F = 1$ to $F' = 2$ transition of the D_2 line, thus effectively transferring population to the $F = 2$ ground-state. The second, right-circularly polarized beam impinges along the cavity axis and drives the Stark shifted transition $F = 2$ to $F' = 3$ transition with $\Delta m_F = +1$. Thus the atomic state is pumped to the coupled state $|\uparrow\rangle$.

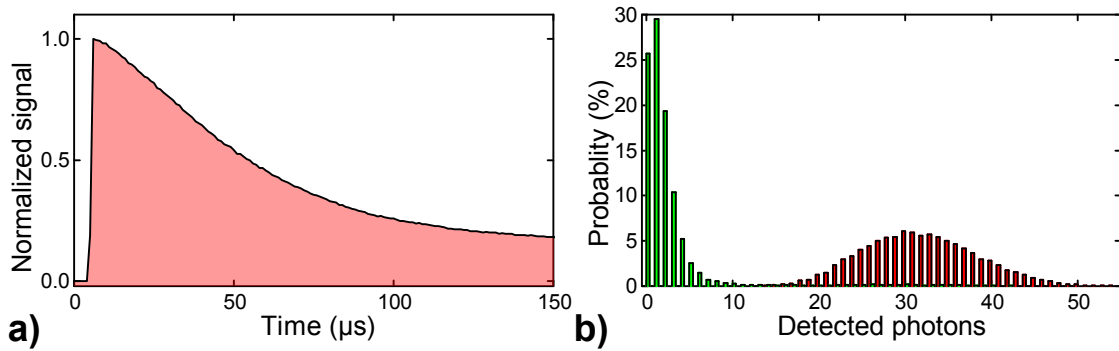


Figure 4.2.: **Optical pumping and state-detection in transmission.** **a)** Typical time trace of normalized resonator transmission for the pumping process to the coupled state $|\uparrow\rangle$. The decay of the signal, which typically saturates after 150 μs , implies a gradually higher success rate of the pumping process. **b)** Atomic state detection in resonator transmission. When the atom is in a detuned state within $F = 1$ (red bars), it is decoupled and the cavity shows a high transmission. The green bars represent the suppressed cavity-transmission signal after the atom was optically pumped to the coupled state. The two well-separated distributions allow for atomic state detection with a fidelity of 98 %. This data will be published in [38].

An averaged cavity-transmission signal for this optical pumping scheme is shown in Figure 4.2a. Since the system is in the strong-coupling regime, transmission of the pumping light is suppressed when the atom is transferred to the desired $|\uparrow\rangle$ state, which causes a decay of the signal, typically saturating after 140 to 150 μs . The transmission signal can be used to determine the atomic state. To demonstrate this, we pump the atom to the far detuned $F = 1$ manifold before applying resonant light along the cavity axis for 10 μs . This signal serves as a reference for the detection of an uncoupled atomic state. Then the atom is pumped to the coupled state and the suppressed cavity transmission is recorded for another 10 μs (see Figure 4.2b). Here, the green (red) bars represent the probability to detect a certain photon number after (before) the optical pumping interval. We observe

two well separated distributions. Setting the threshold of discrimination to 13 detected photons leads to a state detection fidelity of 98 % [36]. This is mainly limited by events in which the atom was scattered out of $|\uparrow\rangle$ by birefringence induced left-circular components in the pumping beam along the resonator. However, no improvement was observed when the pumping duration was increased or when the input polarization of the pumping beam was scanned. The fraction of events in which the atom was pumped into an uncoupled state amounts to 3 % and directly determines the fidelity of the state preparation to be 97 %.

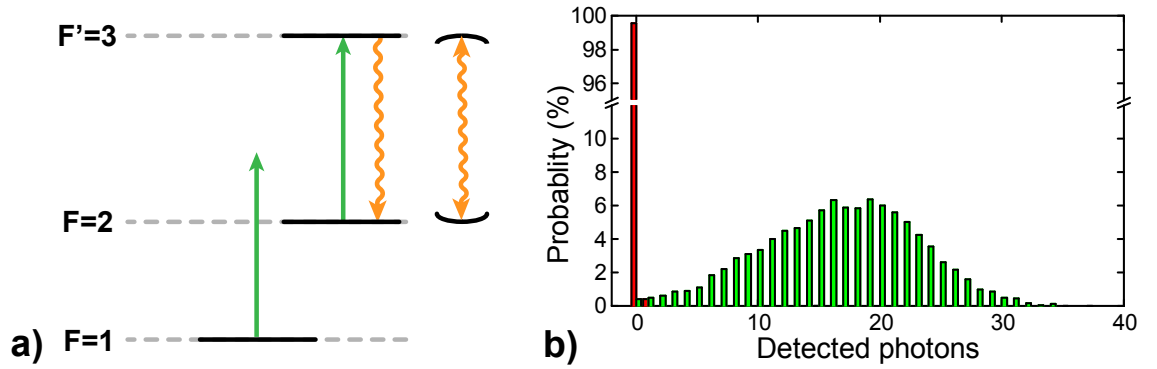


Figure 4.3.: **Cavity enhanced hyperfine-state detection.** **a)** Relevant level scheme for this type of state detection. A laser beam (green arrow), which is resonant with the cavity and the atomic transition $|2,2\rangle \rightarrow |3',3\rangle$ impinges from the side. Fluorescence to the cavity mode is enhanced by the Purcell effect. **b)** Probability to detect a certain number of scattered photons for a detection interval of 3 μ s when the atom is prepared in $F = 1$ (red bars) or $F = 2$ (green bars). Two well separated distributions are observed. This data set will be published in [37].

Another technique for atomic state detection is cavity enhanced fluorescence hyperfine-state detection, which has become a standard technique in our laboratories [77]. Here, no discrimination between coupled and uncoupled states is made but the atomic hyperfine state is detected. To this end, light resonant with the transition $|2,2\rangle \leftrightarrow |3',3\rangle$ is applied from the side (typically for 3 μ s). If the atom is on resonance, i.e. in $F = 2$, then it scatters many photons and the Purcell effect directs them into the resonator mode, thus providing a strong fluorescence signal. As presented in Figure 4.3b, no photon is detected if the atom is pumped to the detuned $F = 1$ manifold (red bars) in 99.6 % of all trials, which is in stark contrast to the number of detected photons if the atom was pumped to the resonant manifold (green bars). In this case, one or more photons are detected in 99.6 % of all trials, leading to a state discrimination fidelity of 99.6 %. Due to its better discrimination fidelity, atomic hyperfine states are detected with this technique in the rest of this thesis.

4.3. State manipulation with Raman lasers

Control over the atomic qubit state is a prerequisite for the successful realization of the storage and read-out protocols. To this end, a pair of Raman beams is employed in our setup. The experimental implementation via spectral addressing of Raman transitions is presented in Section 4.3.1. After presenting data that illustrates the deterministic coupling between $|2, 2\rangle$ and $|1, 1\rangle$, we demonstrate full control over the phase relation of superpositions in Section 4.3.2.

4.3.1. Coupling of hyperfine levels in a guiding magnetic field

The Raman laser beams operate at 795.3 nm close to the Rubidium D_1 line and are generated from a single laser beam which intrinsically facilitates phase stability [78]. The necessary frequency difference of 6.8 GHz is bridged by the generation of sidebands with a fibre-coupled electro-optic modulator. The modulated beam is referred to as the probe beam, while the other is denoted as the control beam. Acousto-optic modulators are used to bring both beams into two-photon resonance. The beams are shifted in opposite frequency directions, which suppresses unwanted interference between the carrier of the modulated and unmodulated beam. The beams are then overlapped on a polarizing beam splitter and impinge onto the atoms from the side of the resonator.

On two-photon resonance, the effective Raman-Rabi frequency between the two ground-states is [78]:

$$\Omega_R = \frac{\Omega_C \Omega_P}{2\Delta} \quad (4.2)$$

Here, Ω_C (Ω_P) denotes the Rabi frequency of the control (probe) beam. When choosing the one-photon detuning Δ one has to consider two points. The detuning has to be sufficiently large to avoid population of the excited state and the resulting projection of the atomic state due to spontaneous emission of a photon of distinguishable frequency. However, all rotations on the atomic qubit have to be fast compared to all other decoherence mechanisms, e.g. fluctuations of the magnetic field. This means that Δ has to be chosen sufficiently small, since the effective Rabi frequency Ω_R is inversely proportional to the detuning and the available laser intensity sets an upper bound to $\Omega_C \Omega_P$. In the light of these considerations, the one-photon detuning was set to $\Delta = -0.15$ THz for the experiments in this work.

Similar to the treatment of Stark shifts in a dipole trap (see Section 4.1), one has to consider the polarization configuration of the laser beams and sum over all possible transitions in order to calculate the effective Raman-Rabi frequency. In our configuration the detuning Δ is small compared to the fine structure splitting of the excited state and large compared to the hyperfine-splittings. Therefore, one can disregard transitions to the $5P_{3/2}$ state, as

their contribution to the effective Rabi frequency is very low. Transitions to the $F' = 1$ and $F' = 2$ hyperfine manifold of the $5P_{1/2}$ state are equally contributing, since the detuning Δ is large compared to their energy difference. Summing over the transitions to $5P_{1/2}$ leads to a suppression of transitions with $\Delta m_F = 0$ and both beams polarized along the quantization axis, expressed as π/π , as well as transitions with $\Delta m_F = \pm 2$. The desired transition, which would couple the qubit states $|\uparrow\rangle \leftrightarrow |\downarrow\rangle$, is one of the allowed transitions with $\Delta m_F = +1$ and a π/σ^+ polarization configuration.

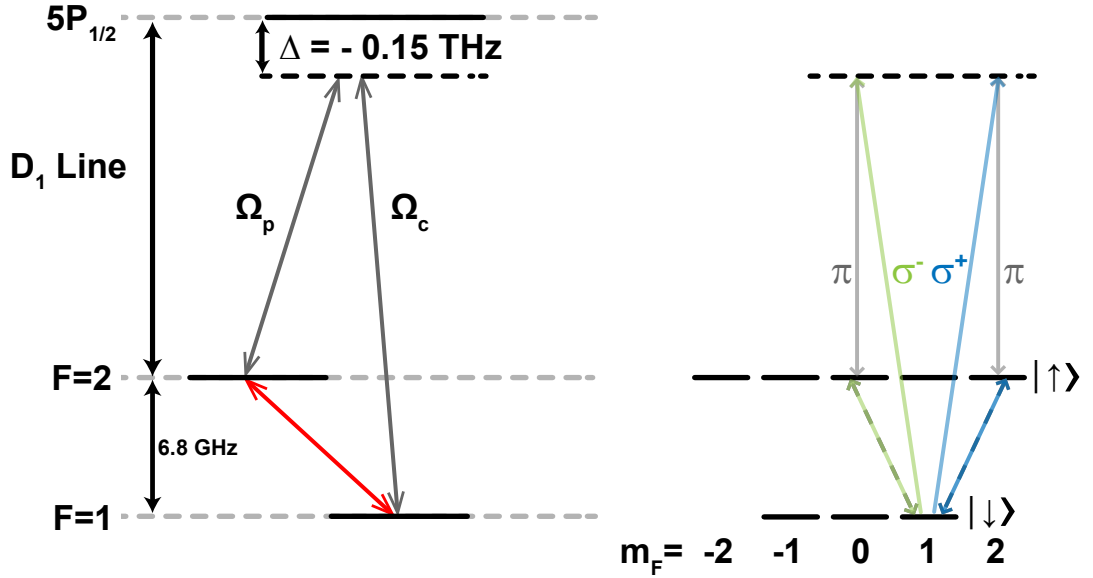


Figure 4.4.: **Raman laser configuration.** **Left:** ^{87}Rb D_1 line. When shining in two light fields on two-photon resonance (grey arrows), an effective coupling between the hyperfine ground-states is realized (red arrow). **Right:** Coupling of the degenerate Zeeman states to the co-propagating Raman beams. In the experiment, one beam (grey/ π) is linearly polarized along the cavity axis, i.e. the quantization axis, thus only driving transitions with $\Delta m_F = 0$. The other beam is orthogonally polarized and is therefore in a balanced polarization superposition of σ^+ (blue) and σ^- (green). This scheme allows for the coupling of the atomic qubit states $|\uparrow\rangle$ and $|\downarrow\rangle$ (dark blue-blue arrow). However, if the Zeeman levels are degenerate, unfavorable transitions are coupled as well and the atomic population is driven out of the qubit subspace (dark green-green arrow).

The coupling of the qubit states can be realized in a co-propagating beam configuration with orthogonal linear polarizations. One of the beams is π -polarized along the quantization axis, i.e. the resonator axis, and the other beam is orthogonally polarized. With regard to the quantization axis, an orthogonal linear polarization can be expressed as a balanced superposition of σ^+ and σ^- . The beams fulfil the aforementioned π/σ^+ polariza-

tion configuration if the σ^+ component of the orthogonally polarized beam is taken into account. Yet, the other polarization component would introduce an undesired coupling of $|\downarrow\rangle$ to $|2,0\rangle$ and rotate atomic population out of the qubit Hilbert space, see Figure 4.4.

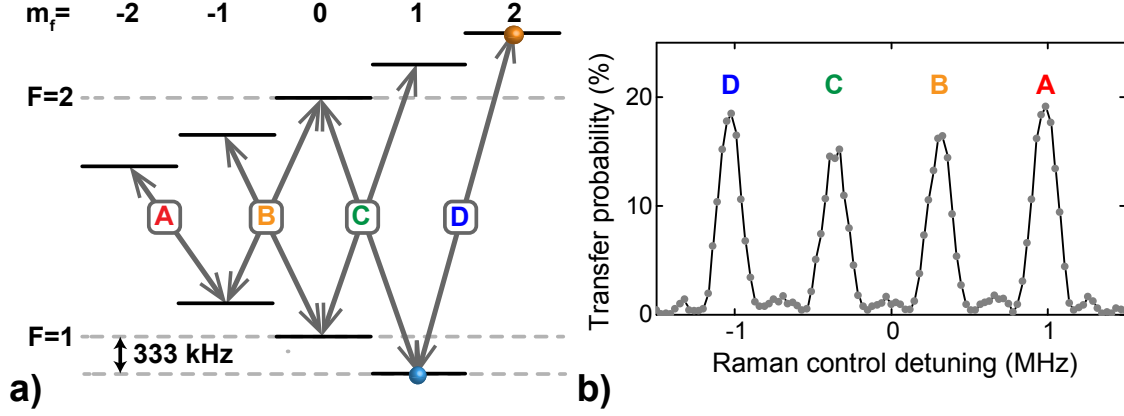


Figure 4.5.: **a)** Level shifts of the ^{87}Rb ground-state in a guiding magnetic field of $B = 0.5 \text{ G}$ along the quantization axis and the expected Raman couplings in a $\pi/\sigma^\pm$ beam configuration (grey arrows). The observed transition frequencies are marked with coloured letters. Thus four distinguishable transition lines are expected in a spectroscopic measurement. The desired coupling of the qubit states (blue and orange spheres) is denoted by **D**. **b)** Measured Raman transition spectrum for the respective magnetic field configuration of panel a). The measured data points are denoted as grey filled circles and connected by a black line for increased visibility. The line broadening and the smaller side-peaks stem from the Fourier spectrum of the Raman pulses. Nomenclature and colour-code correspond to panel a).

In order to suppress the coupling of $|\downarrow\rangle$ with $|2,0\rangle$, a magnetic field of $B = 0.5 \text{ G}$ is applied along the resonator axis. The energy levels of the $5S_{1/2}$ ground-state are split by $\Delta E_{F,m_F} = \mu_B m_F g_F B$, which leads to a splitting of 0.33 MHz between neighbouring energy levels. Here, μ_B denotes the Bohr magneton and $g_F = \pm 0.5$ represents the Landé g-factor of the $F = 1$ (negative sign) or the $F = 2$ (positive sign) hyperfine manifold. The applied magnetic field lifts the degeneracy of the Zeeman states and allows for spectral addressing of specific Raman couplings. In the aforementioned $\pi/\sigma^\pm$ beam configuration one expects six different couplings, where two lines still exhibit frequency degeneracies (see Figure 4.5a). Thus four distinguishable lines are observed in a Raman spectrum, which is recorded by preparing the atom in the $F = 1$ manifold, applying the Raman beams with varying two-photon detuning and employing fluorescence state detection. The resulting spectrum is presented in Figure 4.5b. The observed linewidth originates from the spectral width of the $5 \mu\text{s}$ long rectangular Raman pulses.

The Fourier spectrum of a rectangular pulse with temporal length T , central frequency

ν_0 and amplitude A corresponds to $AT \cdot \text{sinc}[(\nu - \nu_0)T]$ and therefore has a vanishing amplitude at absolute frequency $|\nu - \nu_0| = 1/T$. The implementation of the memory protocol of Section 3.1 will require Raman $\pi/2$ -pulses, which bring the atomic state into a balanced superposition. By choosing a pulse length of $1.7\mu\text{s}$, applied Raman $\pi/2$ -pulses exhibit no spectral contribution at $\pm 0.6\text{MHz}$ which hampers undesired coupling to $|2, 0\rangle$.

The coupling of the qubit states $|\uparrow\rangle$ and $|\downarrow\rangle$ can be demonstrated by the measurement of Rabi oscillations in the qubit subspace. If the $|2, 0\rangle$ state would still couple to $|\downarrow\rangle$ one would observe a beating pattern in the Rabi oscillation due to the different coupling strengths of the transitions. To perform the measurement, the atom is initially prepared in $|\uparrow\rangle$ (see Section 4.2) and the Raman beams are applied for a varying duration. Recording the transfer probability to $F = 1$ via hyperfine-state detection results in the oscillation pattern of Figure 4.6. The damped sine fit to the data gives a visibility of 98 %, which is limited by the fidelities of state preparation and read-out. We find a decay time of 0.2ms , in good agreement with previous experiments in our setup [28].

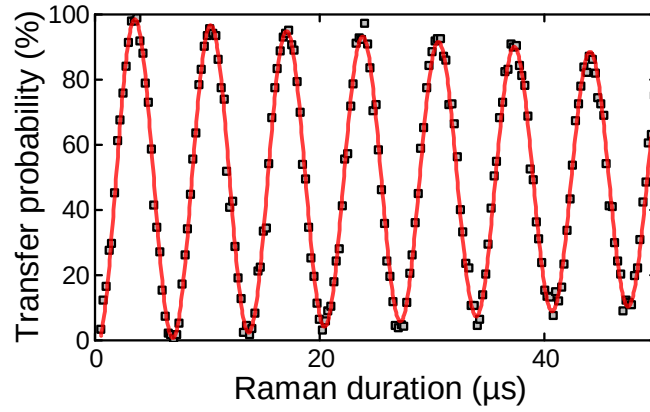


Figure 4.6.: **Rabi oscillations of the atomic state.** The atom is prepared in the $|\uparrow\rangle$ state and the Raman beams are applied for a varying duration. Subsequent hyperfine-state detection returns the transfer probability to the $F = 1$ manifold. The red curve represents an exponentially damped sine fitted to the data, with a visibility of 98 % and a damping time constant of 0.2ms .

4.3.2. Controlled atomic state rotations

The essential steps towards control over the atomic qubit have been presented in the previous section. However, full control over the qubit state requires the ability to prepare any desired superposition $|\psi\rangle = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$, with normalized complex coefficients α and β . The most convenient way of representing such a system is the Bloch sphere picture with a real-valued three-dimensional state vector $\vec{S}$ that represents the complex two-dimensional

qubit state $|\psi\rangle$:

$$|\psi\rangle = \cos\left[\frac{\phi}{2}\right] |\downarrow\rangle + e^{i\theta} \sin\left[\frac{\phi}{2}\right] |\uparrow\rangle \longleftrightarrow \begin{pmatrix} \cos(\theta) \sin(\phi) \\ \sin(\theta) \sin(\phi) \\ \cos(\phi) \end{pmatrix} = \vec{S}. \quad (4.3)$$

Here, the eigenstates of σ_3 , i.e. $|\uparrow\rangle$ and $|\downarrow\rangle$, lie on the poles of the sphere, whereas balanced superpositions lie on the equator. The coordinate system rotates along the z axis with an angular frequency that is equal to $\frac{E_{|\uparrow\rangle} - E_{|\downarrow\rangle}}{\hbar}$. When applying Raman beams the system is generally considered in a rotating frame with an angular frequency equal to the frequency difference of the beams. Figure 4.7a presents the Bloch sphere as a grey transparent sphere and the eigenstates of the Pauli spin matrices, e.g. $\sigma_2 |\downarrow_y\rangle = -|\downarrow_y\rangle$, are highlighted as coloured spheroids and attached to the correspondingly coloured axis label.

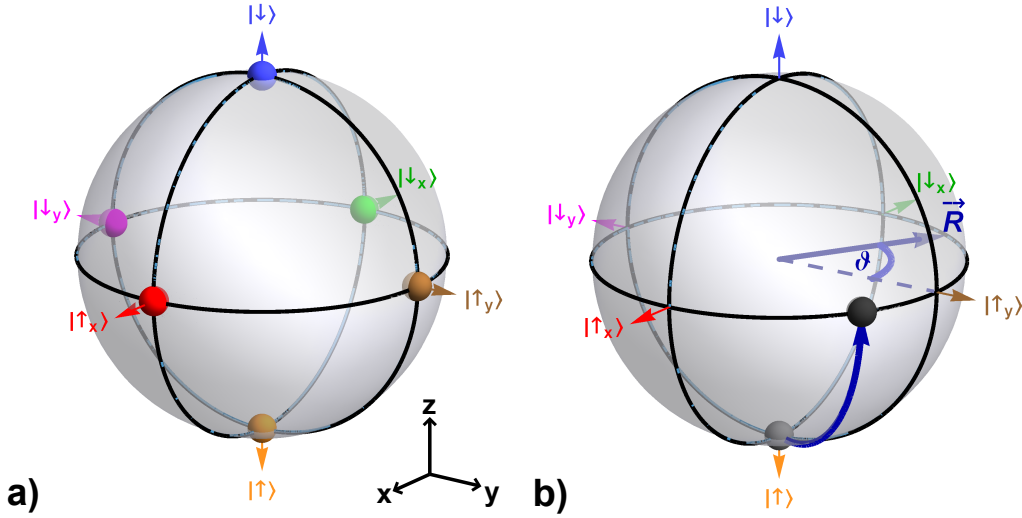


Figure 4.7.: **a)** Bloch sphere representation of the atomic qubit in Stokes space. The labels lie on the respective coordinate axes and are defined as $|\downarrow\rangle \equiv |F=1, m_f=1\rangle$, $|\uparrow\rangle \equiv |F=2, m_f=2\rangle$, $|\downarrow_x\rangle \equiv (|\downarrow\rangle - |\uparrow\rangle)/\sqrt{2}$ and $|\downarrow_y\rangle \equiv (|\downarrow\rangle - i|\uparrow\rangle)/\sqrt{2}$. Coloured spheroids denote the position of the atomic state that corresponds to the axes label of the same colour. **b)** Action of Raman rotations on atomic states. The initial atomic state $|\uparrow\rangle$ (grey sphere) is transferred by $R_{\zeta=\pi/3}(\pi/2)$ (dark blue arrow) to a state in the equatorial plane (black sphere).

Applying Raman beams on two-photon resonance can be described by a rotation of the qubit state around a vector in the equatorial plane. The direction of this vector $\vec{R}$ is given

by the phase difference ζ between the two light fields

$$\vec{R} = \begin{pmatrix} \sin(\zeta) \\ \cos(\zeta) \\ 0 \end{pmatrix}. \quad (4.4)$$

The absolute value of $\vec{R}$ is given by the Raman-Rabi frequency Ω_R , such that the time evolution of the state vector $\vec{S}$ is:

$$\frac{\partial \vec{S}}{\partial t} = \vec{R} \times \vec{S} \quad (4.5)$$

Assuming a constant Rabi frequency applied for a time t results in a pulse area of $\Lambda = \int_{t_0}^{t+t_0} \Omega_R dt' = \Omega_R t$ and the action of the Raman rotation can be described by a rotational matrix $R_\zeta(\Omega_R t)$. As an example, a rotation by $\Lambda = \pi/2$ from $|\uparrow\rangle$ to the state $(|\downarrow\rangle + e^{i\pi/3}|\uparrow\rangle)/\sqrt{2}$, which is equivalent to $R_{\pi/3}(\pi/2)|\uparrow\rangle$ is shown in Figure 4.7b. In the experiment, control over the phase difference is obtained by switching the phase of the RF-signal driving one of the acousto-optic modulators using a direct digital synthesizer (Analog Devices AD9910). In order to increase the readability, we are going to re-write the notation of frequently used rotations, which coincide with a coordinate axis, e.g. $R_0(\Lambda) \equiv R_y(\Lambda)$ and $R_\pi(\Lambda) \equiv R_{-y}(\Lambda)$.

We demonstrate the control over the atomic qubit in a Ramsey spectroscopy measurement. To this end, the atom is prepared in $|\uparrow\rangle$ and rotated to $|\uparrow_x\rangle \equiv (|\downarrow\rangle + |\uparrow\rangle)/\sqrt{2}$ by $R_y(\pi/2)$. After a dark time τ , another $R_y(\pi/2)$ pulse is applied and the atomic state is measured via fluorescence hyperfine-state detection. Under the assumption that the atomic state has not been disturbed during the dark time interval, one expects the atomic state to be transferred to $|\downarrow\rangle$ by the second Raman pulse: $R_y(\pi/2)|\uparrow_x\rangle = |\downarrow\rangle$. Yet the phase of the atomic superposition can change and the atomic state can therefore rotate in the equatorial plane of the Bloch sphere. Possible reasons for a rotation are, for example, a detuning δ from two-photon resonance of the Raman beams, which can be described by a rotation along the z axis with a covered area of $\Lambda = \delta \cdot \tau$. This rotation occurs because the rotating frame is chosen with a frequency offset of δ . We therefore vary the frequency of one Raman beam and record an oscillation pattern with expected periodicity $1/\tau$. Another possible disturbance are magnetic field fluctuations and therefore fluctuations in the qubit energy difference $E_{|\uparrow\rangle} - E_{|\downarrow\rangle}$. When these fluctuations are random, the acquired phase is also random, and the visibility of the oscillation pattern is reduced.

Figure 4.8a presents the transfer probability from $|\uparrow\rangle$ to $|\downarrow\rangle$ in a Ramsey sequence with the last pulse having no phase difference (black) or a phase difference of $\pi/2$ (red data) with respect to the first pulse. The oscillation period for both curves is the same and corresponds to a dark time of $\tau = 7.5 \mu\text{s}$. The fact that the red curve is phase shifted by $\pi/2$ with respect to the black data demonstrates the experimental control over the rotation axis via the phase relation of the Raman lasers. Hence, all prerequisites for the implementation of

the storage and read-out protocol are fulfilled. Both curves exhibit a visibility of 95(1) %, limited by the preparation fidelity, errors in the pulse area, decoherence and state-detection fidelity of the atomic qubit. Both curves are offset to a lower transition probability by 1.3(1) %. This hints at imperfect state preparation in $|\uparrow\rangle$, which would generate a constant background of events with the atom being in $F = 2$ and therefore shift all data to lower transfer probabilities.

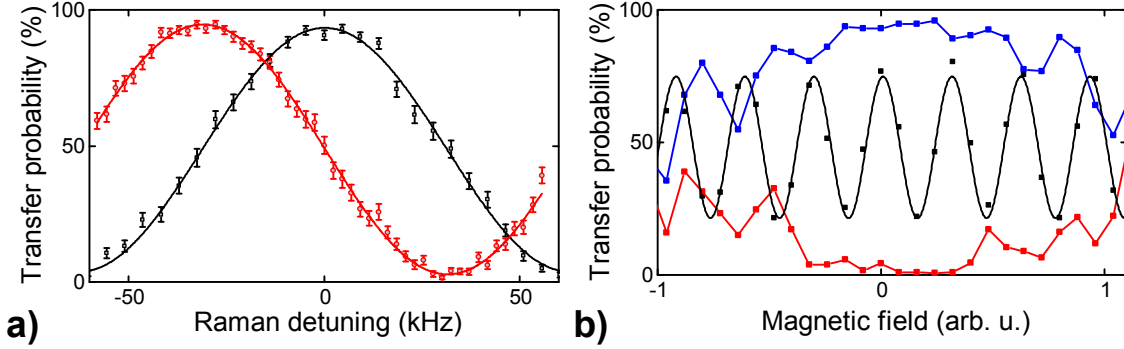


Figure 4.8.: **Demonstration of full control over the atomic qubit state.** a) Ramsey spectroscopy for varying two-photon detuning. Black (red) data corresponds to a standard (phase shifted) Ramsey sequence. Fits to both data sets return the same visibility of 95 % and the same dark time $\tau = 7.5 \mu\text{s}$. This data set will be published in [37]. b) Demonstration of spin echo for a dark time of $\tau = 20 \mu\text{s}$ when varying the guiding magnetic field along the cavity axis. The black data shows the result of a standard Ramsey sequence and the black curve is a sine fitted to the data. For the red (blue) data set, a rotation $R_y(\pi)$ ($R_x(\pi)$) is applied after half the dark time has passed. This causes the atomic state to be in a well-defined superposition state at the end of the sequence which is read out with high visibility.

The randomization of the phase relation of atomic superpositions might be an obstacle for experiments which require long coherence times. When extending the dark time interval τ of the previous Ramsey sequence the visibility of the oscillations is expected to be decreased because the impact of random noise in the two-photon detuning is increased by larger τ . The previously demonstrated phase sensitive addressing of the atomic state might be crucial in overcoming this obstacle.

In the following, the Ramsey sequence is extended to a longer dark time of $\tau = 20 \mu\text{s}$ and the two-photon detuning is varied by scanning the magnetic field along the quantization axis, i.e. by varying the current applied to the coils which generate the field. The black data in Figure 4.8b represents the expected oscillation with a damped visibility when compared to panel a). The sine fitted to the data shows an offset to lower transfer probabilities of 2(1) %. The red and blue data sets of Figure 4.8b are explained in the following.

By employing pulse sequences which have been pioneered in experiments with nuclear spins [79], one can transfer the atomic spin state into a well-defined superposition state at a given point in time. In order to demonstrate this, the Ramsey sequence is extended to a spin echo sequence by applying a π -pulse after half of the dark time τ has passed. Under the assumption that the two-photon detuning remains constant on the time scale of τ , the sequence can be expressed as: $R_y(\frac{\pi}{2}) R_z(\frac{\tau \cdot \delta}{2}) R_y(\pi) R_z(\frac{\tau \cdot \delta}{2}) R_y(\frac{\pi}{2})$. The atomic state is initialized in a balanced superposition state $R_y(\frac{\pi}{2}) |\uparrow\rangle = |\uparrow_x\rangle$, which acquires a phase due to two-photon detuning during the first half of the dark time interval. The following $R_y(\pi)$ rotation reverses the phase $\phi = \tau \cdot \delta/2$ of the atomic state and imprints an additional phase shift of π

$$R_y(\pi) \left(|\downarrow\rangle + e^{i\phi} |\uparrow\rangle \right) / \sqrt{2} = \left(|\downarrow\rangle - e^{-i\phi} |\uparrow\rangle \right) / \sqrt{2}.$$

Therefore one expects that the atomic superposition is rotated into $|\downarrow_x\rangle$ after the dark time interval, independent of the exact two-photon detuning. For this type of sequence, the expected transfer probability is low as the final Raman $\frac{\pi}{2}$ -pulse is used to read out the atomic superposition and thus rotate the state back into the initial state: $R_y(\pi/2) |\downarrow_x\rangle = |\uparrow\rangle$. The red data in Figure 4.8 b) employs this type of sequence. As expected, a low average transfer probability (3 %) around two-photon resonance is observed which clearly surpasses the visibility of the usual Ramsey sequence (black data). The blue data corresponds to an alternated spin echo sequence with a phase-shifted $R_x(\pi)$ rotation which causes the atomic state to rotate into $|\uparrow_x\rangle$ after the dark time has passed. Thus, leading to a high average transfer probability (94 %) around two-photon resonance. The discrepancy between maximum (blue) and minimum (red) transfer probabilities is within the error of the offset of the sine fitted to the black data set and might be caused by imperfect state preparation.

This experiment presents phase sensitive addressing of atomic superpositions as a tool for new experimental pathways in this system. In the future, optimal control [80], universal dynamical decoupling [81] and composite pulses [82] might be implemented to improve the robustness of the system against sources of random noise.

5. Heralded quantum state mapping between single atoms and photons

The previous chapters have introduced the required experimental control over a single atom and the theoretical considerations for the implementation and characterization of the atom-photon interaction mechanism.

This chapter will present the main experimental results of this work. In Section 5.1, the action of a coupled atom on reflected photons as a function of their spectral distribution or their central frequency is investigated. The experimental setup and the temporal sequence are described in Section 5.2. Heralded storage of the polarization of a photon in the spin of an atom is performed in Section 5.3. Additionally, the reverse process, i.e. the transfer of an atomic spin state onto the polarization state of a photon, is demonstrated in Section 5.4. The chapter is concluded with a demonstration of the versatility of the interaction mechanism, by the generation of an atom-photon entangled state in Section 5.5.

5.1. Frequency dependence of the atom-photon interaction mechanism

Theoretical considerations, in Section 3.3.2, revealed the importance of resonance between the frequency of the atomic transition, the resonator and the impinging photon in order to achieve the ideal interaction with a phase shift of π . Moreover, these considerations assumed a photon with narrow spectral distribution and therefore broad temporal envelope. This section experimentally investigates the dependence of the interaction mechanism on the frequency bandwidth and frequency detuning of impinging photons.

The impact of frequency bandwidth is investigated by preparing the atom in the coupled $|\uparrow\rangle$ state, reflecting a faint linearly polarized laser pulse off the cavity and detecting the reflected pulse in the same linear polarization basis. From the previous treatment (see Section 3.2), one expects to observe a polarization flip to the orthogonal polarization if the spectral distribution of the pulse is sufficiently narrow. Figure 5.1 shows the polarization flip probability $p(|D\rangle \rightarrow |A\rangle) = p_{|A\rangle} = \frac{N_{|A\rangle}}{N_{|D\rangle} + N_{|A\rangle}}$ depending on the full width at half maximum (FWHM) of a reflected Gaussian pulse. The polarization flip probability remains constant for photons which are longer than 0.5 μs , here the interaction mechanism works as

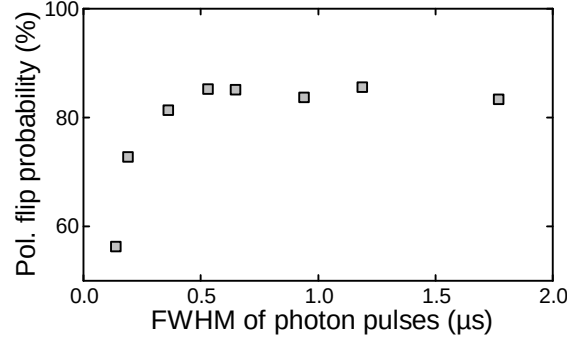


Figure 5.1.: **Bandwidth of the interaction mechanism.** The data points show the probability of observing an orthogonal linear polarization state after reflection of photons with varying FWHM of the Gaussian temporal envelope. The flipping probability remains constant for photon lengths above $0.5\mu\text{s}$ and decreases for shorter photons. This data set is also published in [38].

intended. Pulses with a broader spectral distribution experience varying phase shifts and therefore the flip probability decreases rapidly for pulses that exhibit a shorter temporal envelope

The validity of the theoretically predicted phase shift given in Figure 3.5 can be verified by shifting the central frequency of an impinging $|D\rangle$ -polarized pulse with respect to the fixed atomic and cavity frequencies. Figure 5.2a presents the measured polarization fractions in the detection bases, R/L, D/A and H/V, as a function of the frequency detuning of a reflected pulse with a narrow spectral distribution. The blue data points correspond to the right-circular polarized fraction $p_{|R\rangle} = \frac{N_{|R\rangle}}{N_{|R\rangle} + N_{|L\rangle}}$. As indicated by the reflection spectra in Section 3.3.2, a balanced reflectivity for coupled $|R\rangle$ -polarized components and uncoupled $|L\rangle$ -polarized components of the pulse is observed on resonance. The fraction of right circular components decreases for larger absolute detuning due to the higher absorption of the system on a normal mode. The black data points correspond to the polarization flip probability $p_{|A\rangle}$ which stays almost constant around resonance and falls off rapidly for larger detuning. Finally, the red data points present the fraction of $|H\rangle$ -polarized light $p_{|H\rangle} = \frac{N_{|H\rangle}}{N_{|V\rangle} + N_{|H\rangle}}$ which undergoes a large amplitude change when scanned over the resonance.

One can calculate the phase shift between right-circular and left-circular polarization components via $\text{Arg}\left(\frac{p_{|H\rangle} - p_{|V\rangle}}{p_{|D\rangle} - p_{|A\rangle}}\right)$. Figure 5.2b finds excellent agreement between experimentally observed data and the red curve, which is derived from the previous theoretical treatment (Section 3.3.2). The theoretical curve had to be shifted by a slight offset of $+0.2\text{MHz}$ to match the data. This offset might be caused by temperature related drifts of the cavity resonance and the fact that the cavity is locked in a Pound-Drever-Hall scheme [83] to the polarization eigenmode close to $|A\rangle$ which is 0.4MHz detuned from the orthogo-

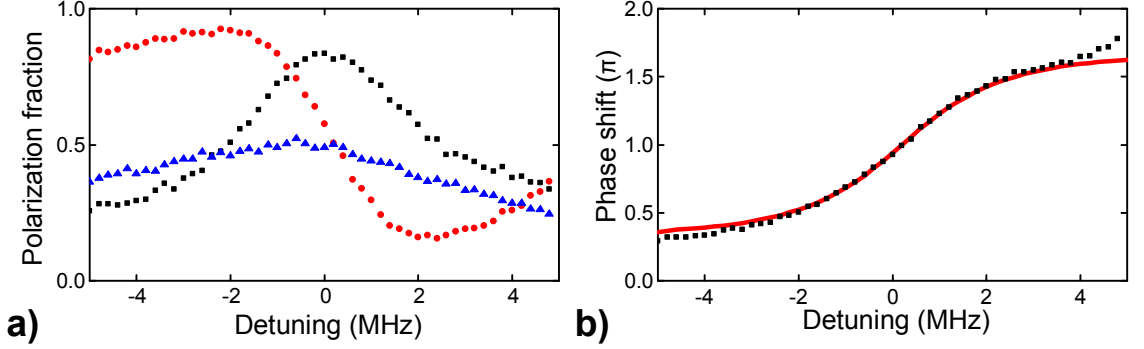


Figure 5.2.: **Investigation of the experimental phase shift.** **a)** Measured polarization fraction as a function of frequency detuning of a $|D\rangle$ -polarized impinging laser pulse if the atom is prepared in the coupled state. The normalized polarization fraction of the reflected pulse is given for $p_{|A\rangle}$ (black data points), $p_{|H\rangle}$ (red data points) and $p_{|R\rangle}$ (blue data points). **b)** Comparison of theoretically expected (red) and measured phase shift (black data points) between left-circular and right-circular polarization which is corresponding to the phase difference between the uncoupled and the coupled reflection event. The parameters of the theoretical curve are deduced from the input-output treatment detailed in Section 3.3.2.

nally polarized eigenmode (see Section 3.3.3 for further information on birefringence in the system).

These measurements underline the importance of the spectral distribution of the impinging pulse, as suggested in the original proposal of this mechanism [35], and the long and short term stability of the setup.

5.2. Experimental setup & storage sequence

Experimental setup

To mimic an independent single-photon source, the atom-photon interaction is characterized with weak coherent pulses, which contain less than one photon on average. The resulting low signal level requires the best possible photon-detection efficiency, improving the performance of our setup in three ways. First, the signal-to-noise ratio, i.e. the detection rate of a photon versus detector dark count rate, is increased. Second, increased detection probabilities result in a faster data acquisition and simplify the identification of systematic drifts over time. Third, photons serve as heralds for the storage process, thus a higher detection efficiency leads to a higher storage efficiency. In previous experiments, detection setups were fibre coupled to the atom-cavity system(s) [28, 29, 54], mainly to

suppress stray light. However, the non-unity coupling of light into an optical fibre and its intrinsic losses reduce the detection efficiency in this case by a factor of ~ 0.9 .

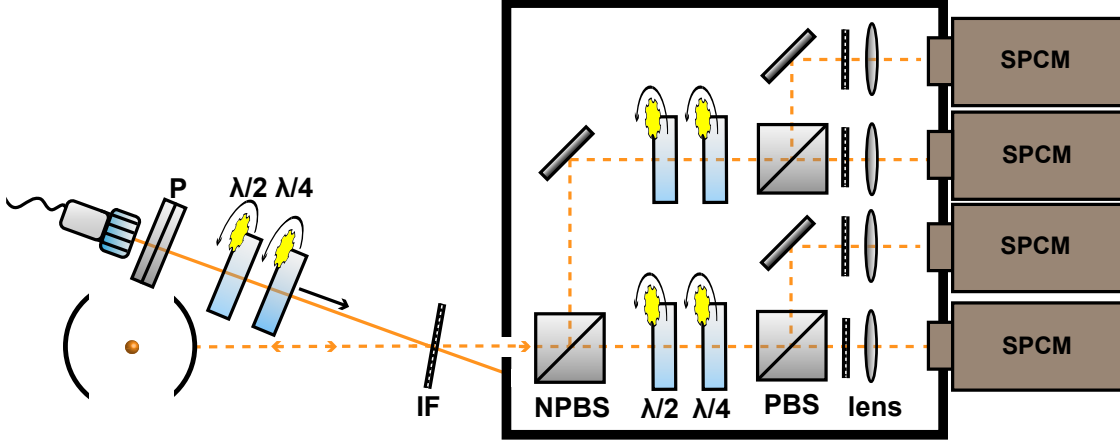


Figure 5.3.: **Schematic of the free-space detection setup.** A faint beam of resonant light at 780 nm is reflected from an infrared filter (IF) and directed onto the atom-cavity system. The polarization of the impinging beam is set by a polarizer (P) and a combination of half-wave and quarter-wave plates mounted on rotation stages ($\lambda/2$ and $\lambda/4$). Light, which is reflected off the atom-cavity system, passes the infrared filter and enters a light-tight enclosure (depicted as a black rectangle in the schematic) with a built-in polarization-resolving detection setup. Here, the beam is split by a non-polarizing 50 : 50 beam splitter (NPBS) and motorized waveplates in front of a polarizing beam splitter (PBS) define the detection basis of the setup. After passing through additional infrared filters, the light beam is focused onto single-photon counting modules (SPCM), which are connected to a time-to-digital converter for data acquisition.

Figure 5.3 presents a schematic of the free-space detection setup used in this work. A resonant laser pulse is sent onto an infrared band-pass filter (Semrock LL 780) which is slightly tilted and mimics a non-polarizing beam splitter with low reflectivity. About 98% of the incident intensity is transmitted and then absorbed by an aperture. The remaining faint reflection impinges onto the atom-cavity system. The reflection off the cavity passes the infrared filter again and enters our detection setup. Quarter-wave and half-wave plates, which are mounted on rotational stages, in front of polarizing beam splitters set the detection bases. The polarization extinction ratios of our polarizing beam splitters are measured to be 1 : 100 in reflection and 1 : 1000 in transmission. The overall detection efficiency of a single photon, reflected from the cavity, is dominated by the quantum efficiency of our SPCMs and determined to be $\eta = 56(5)\%$.

Experimental sequence for polarization storage

In this section, the experimental sequence used to perform the storage experiment is explained. It consists of several individual steps of atomic manipulation (see Chapter 4) and the reflection of a faint laser pulse. The resulting sequence allows for the realization of the storage protocol and is only slightly modified for the read-out and entanglement experiments in Section 5.4 and Section 5.5. The physical implementation and the temporal arrangement of laser pulses as logical blocks is presented in Figure 5.4a and b respectively. The sequence is repeated at a rate of 1 kHz and the recorded digital-data files are composed of 1000 individual runs.

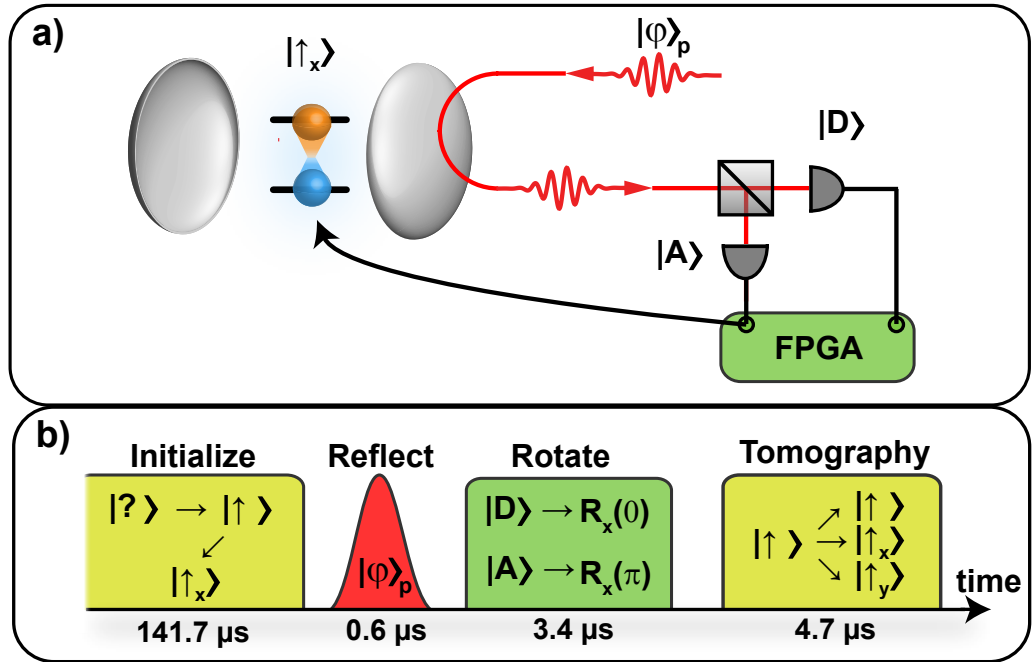


Figure 5.4.: **Implementation of the storage protocol.** a) Physical implementation of the heralded polarization storage protocol. A photon in an unknown polarization state $|\varphi\rangle_p$ is reflected off an atom-cavity-system in the strong coupling regime. After its reflection, the photon is measured in the D/A polarization basis. Depending on the measured polarization the atomic state is either projected onto $|\varphi\rangle_a$ or $\sigma_1 |\varphi\rangle_a$. The measurement result is fed forward to a sequence controller (FPGA), which performs a conditional single-qubit operation and rotates the atomic state into $|\varphi\rangle_a$. b) Temporal sequence of the individual steps needed to perform and characterize the storage experiment. Yellow rectangles display atomic-state manipulation. The red Gaussian shape corresponds to the reflection of a faint laser pulse and the green rectangle represents the conditional atomic state rotation.

Initialization of the atomic qubit

The sequence begins with the initialization of the atomic state in a well defined electronic state. To this end, optical pumping is applied for $140\text{ }\mu\text{s}$ and the atom is transferred from an unknown state $|?\rangle$ into the coupled $|\uparrow\rangle$ state (see Section 4.2).

We exploit the fact that resonant optical pumping along the cavity axis allows for evaluation of the state of the system and apply filters to the data, such that only single atom data is taken into account. The initial two filters act on whole data files which are composed of several single sequence runs.

First, multiple-atom events would falsify results in the hyperfine-state detection. The probability that one of multiple atoms is initially in the coupled state is increased and we therefore observe a suppression of initial cavity transmission for these events. Hence, digital-data files with on average less than $\zeta_{multi} = 16.5$ detected photons during the first $10\text{ }\mu\text{s}$ of the pumping interval are discarded.

Second, atom loss during data acquisition will leave the resonator empty and hinder the characterization of the interaction mechanism. In this case, there is no reduction in cavity transmission at the end of the pumping interval. Hence, we set an upper bound of $\zeta_{loss} = 8$ on the last $10\text{ }\mu\text{s}$ of the pumping interval. Data files which have on average more than eight detected photons in this interval are discarded.

Third, as discussed in Section 4.2, cavity birefringence might result in a left-circularly-polarized pumping light fraction, which pumps the atom into undesired Zeeman states. We therefore employ a third filter on single runs within data files which are verified as single atom events by the aforementioned conditions. Detecting less than $\zeta_{|\uparrow\rangle} = 3$ photons in transmission during the last $10\text{ }\mu\text{s}$ of the optical pumping interval yields a high probability of the atom being in the coupled $|\uparrow\rangle$ state. This leads to a fraction of 44 % when comparing all single atoms runs with the single atom runs that fulfil the condition $\zeta_{|\uparrow\rangle}$. These filtering conditions are summarized in a triplet:

$$\{\zeta_{multi}, \zeta_{loss}, \zeta_{|\uparrow\rangle}\} = \{16.5, 8, 3\}. \quad (5.1)$$

In order to prepare the atom in the desired superposition state $|\uparrow_x\rangle$, the atomic state is rotated by a $1.7\text{ }\mu\text{s}$ long Raman $\frac{\pi}{2}$ -pulse: $R_y(\frac{\pi}{2})|\uparrow\rangle = |\uparrow_x\rangle$. (see Section 4.3.2)

Reflection of a faint laser pulse

After preparing the atom in the correct superposition state, a faint laser pulse is impinging on the atom-cavity system. The spectral distribution of the incident pulse should be within the bandwidth of the interaction mechanism and should therefore have a sufficiently long temporal envelope. Short pulses are yet advantageous to limit temporal dephasing of

the atomic superposition. Considering the discussion of the frequency bandwidth of the interaction mechanism (see Section 5.1), we choose a Gaussian pulse shape with a FWHM of $0.6\,\mu\text{s}$ which satisfies both requirements. The detected mean photon number is 0.03.

Conditional state rotation

According to the storage scheme of Section 3.2 two outcomes for the atomic state are possible. If the photon is detected in $|D\rangle$ then the atomic state is projected into the desired state $|\varphi\rangle_a$. If the polarization state is measured to be $|A\rangle$ then the atomic state is projected into a rotated state $\sigma_1 |\varphi\rangle_a$, which can be brought to the desired state by applying a rotation $R_x(\pi)$. To this end, the signals of all SPCMs are split into two signal lines, one line serving the time-to-digital converter for data acquisition, while the other line is fed to the fast sequence controller, which is a field programmable gate array (FPGA). The FPGA then processes the detected polarization and either does or does not trigger the rotation $R_x(\pi)$. This type of feedforward can also be switched off for characterization purposes.

Quantum-state tomography

The atomic state has to be characterized using quantum state tomography. To this end, the atomic state is rotated into one of three mutually unbiased bases by employing suitable Raman rotations ($R_y(\frac{\pi}{2})$, $R_{-x}(\frac{\pi}{2})$, $R_x(0)$) and subsequent cavity enhanced hyperfine-state detection within $3\,\mu\text{s}$ (see Section 4.2). In order to acquire data on all combinations of detection bases and input states, the atomic detection basis and the input polarization are changed whenever a new atom is loaded.

5.3. Heralded polarization storage

In this section, arbitrary polarization states $|\varphi\rangle_p$ are stored in the atomic state as explained in Section 3.2. Upon detection of a reflected photon, the atomic spin state is projected into an equivalent superposition $|\varphi\rangle_a$, or a rotated version $\sigma_1 |\varphi\rangle_a$, which depends on the measured polarization of the photon. Adding conditional state rotation of the atomic state allows for a heralded storage of polarization that is in principle deterministic. The ideal mapping of the polarization onto the corresponding atomic spin qubit is shown in Figure 5.5.

First, the efficiency of heralding a successful storage event is evaluated. Second, the fidelity of the process is investigated. Finally, the storage process is further analysed by investigating the fidelity of the bare mechanism, i.e. without applying the conditional rotation.

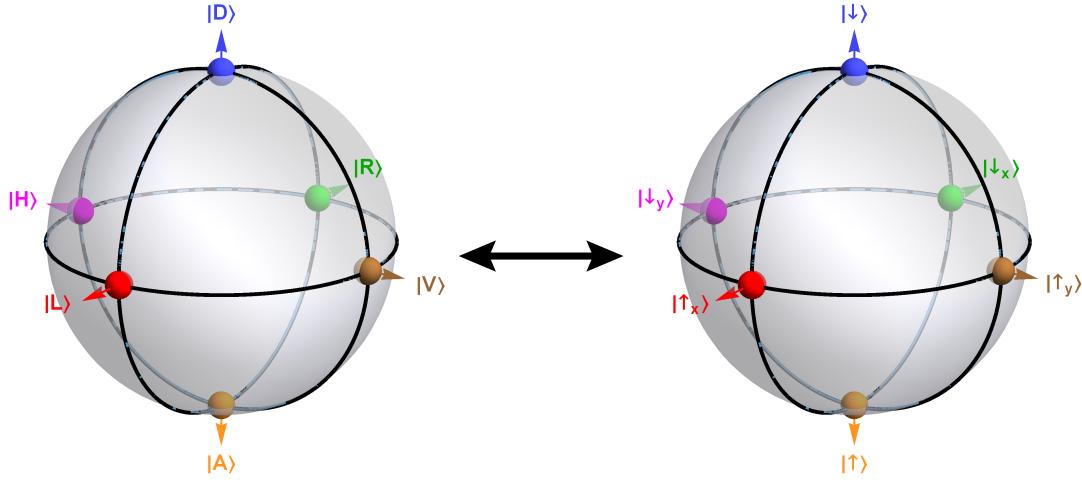


Figure 5.5.: **State representation of the photonic polarization qubit and the atomic spin qubit in Stokes space.** Coloured spheres denote the state of the plot label with corresponding colour. An ideal mapping process would transfer one of the shown polarization states onto the correspondingly coloured atomic state, e.g. $|H\rangle \rightarrow |\downarrow_y\rangle$

Efficiency

The detection of a photon heralds the storage process. Thus, the efficiency of the storage process is given by the probability to detect a photon reflected from the cavity. This probability has two major contributions: the reflectivity of the atom-cavity system and the photon detection efficiency. The cavity-reflection measurements in Figure 3.3 and Figure 3.4 yield the following reflectivities on resonance: If the atomic state is $|\downarrow\rangle$, it is not coupled to the resonator and the reflectivity of the system is 70(2) %, regardless of the polarization impinging on the system. Reflecting an $|R\rangle$ -polarized photon while the atom is prepared in the coupled state $|\uparrow\rangle$ results in a reflectivity of 66(2) %. For the storage protocol the atom is in the state $|\uparrow_x\rangle = (|\downarrow\rangle + |\uparrow\rangle)/\sqrt{2}$, resulting in an overall reflectivity of 69(2) %. As explained in Section 5.2, the quantum efficiency of the detection setup is 56(5) %.

The overall efficiency can be calculated by multiplying the photon survival probabilities for these independent loss channels, which gives 39(4) %. If no conditional state rotation is applied, the efficiency is decreased by a factor of two, because only detection of a $|D\rangle$ -polarized photon yields the correct atomic state.

Fidelity of the storage process

For the reconstruction of both process matrices, the six eigenstates of the Pauli spin operators are used as input polarizations: $\{|H\rangle, |V\rangle, |D\rangle, |A\rangle, |R\rangle, |L\rangle\}$. For each of these input polarizations, the atomic state is obtained by quantum state tomography and the resulting single particle density matrices are used for process tomography via the maximum likelihood technique (see Appendix A). The measured fidelities for each of the six input states are computed using equation (2.7). Here, the input polarizations are assumed to be ideal.

The average state fidelity for heralds in $|D\rangle$ ($|A\rangle$) is $\overline{\mathcal{F}}_{|D\rangle} = 84(1)\%$ ($\overline{\mathcal{F}}_{|A\rangle} = 87(1)\%$). The average state fidelity of the combined process is thus $\overline{\mathcal{F}} = 86(1)\%$. Figure 5.6 presents the action of the storage process on all pure input-states as an ellipsoid in Stokes space.

Comparing the fitted process with the ideal mapping process with $\chi_{00} = 1$, yields a process fidelity of $F_{proc} = 78.7\%$ [see equation (2.9)]. Computing the average state fidelity with the obtained process fidelity and equation (2.10) leads to an average state fidelity of $\mathcal{F} = 85.8\%$. The predicted and measured average state fidelity are in good agreement. Due to cavity birefringence and the averaging over both cavity eigenmodes, the optimum fidelity is found for a slightly phase-rotating process, with an angle of 0.06π and an average state fidelity of 86.2% . The rotation angle is in perfect agreement with the angle expected from the birefringence of our cavity (see Section 3.3.3).

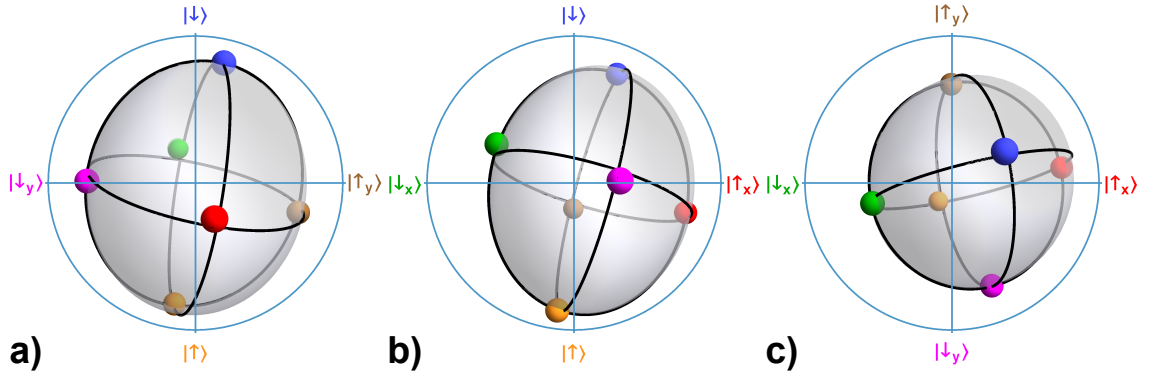


Figure 5.6.: **The combined storage process in the Bloch sphere representation.** The coloured spheres give the measured atomic states for different input polarizations. Sub-figures {a), b), c)} correspond to points of view along $\{\sigma_1, \sigma_2, \sigma_3\}$.

The boundary in average state fidelity for a perfect classical device, which tries to mimic the storage process, can be calculated by taking the mean number of impinging photons $\bar{n} = 0.086$ and the efficiency for a successful process into account [28]. It is assumed that the classical device detects all higher photon number contributions in the coherent input

pulse with unity efficiency and thus infers additional information of the state. Furthermore, the device mimics the success probability of the process and only returns a stored state in 39 % of the trials, even though it is able to detect every single event. Building on previous considerations in our group, one calculates a classical threshold for the average state fidelity of 67.5 % [43]. This value is very close to the simplest assumption of 2/3, because the input photon number is so low and therefore only slightly more information about the state can be inferred by taking higher photon numbers into account.

In summary, we have demonstrated heralded storage of the polarization of a single photon in the spin state of a single atom with a storage efficiency of 39(4) % and an average state fidelity of 86 %. The utilization of conditional single-qubit operations renders the scheme deterministic and the measured fidelities clearly surpass the classical boundary. We point out efficiency could be close to unity with a better herald detection efficiency and a higher reflectivity on resonance.

Fidelity of the bare processes

To characterize the underlying process and its fidelity, we do not apply the feedforward and therefore evaluate the storage process individually for the measurement outcome $|D\rangle$ and $|A\rangle$. Conditioned on detection of the herald in $|D\rangle$ ($|A\rangle$), the mean fidelity of the measured density matrices is $\overline{\mathcal{F}}_{|D\rangle} = 85(2) \%$ ($\overline{\mathcal{F}}_{|A\rangle} = 89(1) \%$), which is within the errors of the previous measurement. Both values are clearly surpassing the classical boundary of 67 %.

Input polarization	Herald in $ D\rangle$	Herald in $ A\rangle$	average
$ H\rangle$	87(4) %	86(3) %	86.5 %
$ V\rangle$	83(5) %	85(3) %	84 %
$ D\rangle$	82(5) %	100(1) %	91 %
$ A\rangle$	98(1) %	85(2) %	91.5 %
$ R\rangle$	75(3) %	80(3) %	77.5 %
$ L\rangle$	84(4) %	95(3) %	89.5 %
avg. fidelity $\overline{\mathcal{F}}$	$\overline{\mathcal{F}}_{ D\rangle} = \mathbf{85(2) \%}$	$\overline{\mathcal{F}}_{ A\rangle} = \mathbf{89(1) \%}$	87 %

Table 5.1.: Fidelities of measured atomic states with the expected ideal states after reflection of one of six input polarizations. The quoted errors are calculated via the Monte-Carlo technique (see Appendix B).

In order to perform process tomography, all six measured output density matrices are considered. The process matrices $\chi_{|D\rangle}$ and $\chi_{|A\rangle}$ are reconstructed in the basis of the Pauli-spin operators using the maximum likelihood technique. A simple fit algorithm would

return non trace-preserving process matrices. But the probability operators $\mathcal{P}_{|D\rangle}$ and $\mathcal{P}_{|A\rangle}$ turn out to be uncorrelated. The reduced eigenvalues λ_2 as well as the respective eigenvectors do not return the same magnitude or direction in Stokes space.

$$\text{Herald in } |D\rangle : \quad \mathcal{P}_{|D\rangle} = \begin{pmatrix} 0.91 & -0.04 - 0.14i \\ -0.04 + 0.14i & 0.76 \end{pmatrix}$$

$$\text{Herald in } |A\rangle : \quad \mathcal{P}_{|A\rangle} = \begin{pmatrix} 0.81 & -0.04 - 0.01i \\ -0.04 + 0.01i & 0.992 \end{pmatrix}$$

$$\begin{aligned} |\lambda_2\rangle_{|D\rangle} &= \begin{pmatrix} -0.15 - 0.51i \\ -0.85 \end{pmatrix} & \lambda_2 &= 0.66 \\ |\lambda_2\rangle_{|A\rangle} &= \begin{pmatrix} +0.96 + 0.20i \\ +0.20 \end{pmatrix} & \lambda_2 &= 0.80 \end{aligned}$$

A credible result for a non trace-preserving process would return vectors which point in the same direction, yet the angle between both estimated eigenvectors in Stokes space is 0.58π . Therefore, we conclude that our process should be trace preserving and use an algorithm that fits only trace-preserving processes to measured states.

The action of both processes are shown as ellipsoids in Stokes space in Figure 5.7. Reconstruction of the process matrix $\chi_{|D\rangle}$ results in:

$$\chi_{|D\rangle} = \begin{pmatrix} 0.784 & 0.025 - 0.061i & 0.014 + 0.081i & -0.031 + 0.135i \\ 0.025 + 0.061i & 0.032 & 0.012 + 0.031i & 0.024 + 0.014i \\ 0.014 - 0.081i & 0.012 - 0.031i & 0.063 & 0.036 - 0.025i \\ -0.031 - 0.135i & 0.024 - 0.014i & 0.036 + 0.025i & 0.121 \end{pmatrix} \quad (5.2)$$

The dominant entry of this matrix is $\chi_{|D\rangle,00} = 0.784$, which corresponds to a direct storage process with fidelity $F_{proc} = 78.4\%$ and an average state fidelity of $\overline{\mathcal{F}}_{|D\rangle} = 85.6\%$. The amplitude of χ_{30} , χ_{03} and χ_{33} indicates a slight phase rotation of $R_z(0.12\pi)$. The expected rotation from birefringence of the two cavity polarization eigenmodes (see Section 3.3.3) matches well with the measured rotation. Hence, the full phase shift due to birefringence is imprinted on $|D\rangle|\downarrow\rangle \rightarrow e^{i\xi}|D\rangle|\downarrow\rangle$.

The reconstructed process matrix for heralds in $|A\rangle$ is:

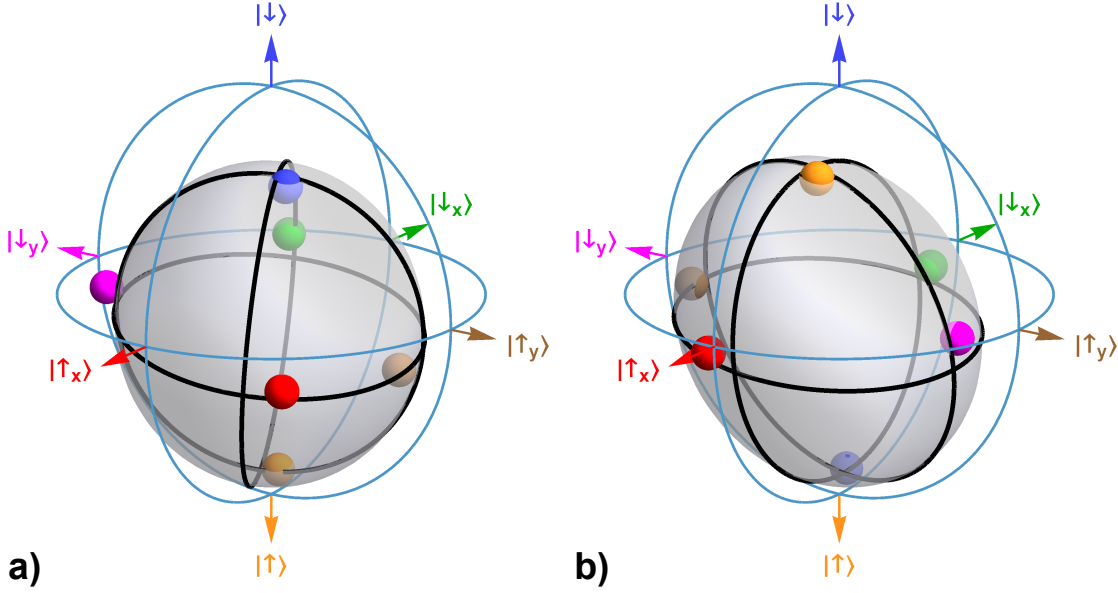


Figure 5.7.: **Storage processes without feedforward.** Representation of the processes $\chi_{|D\rangle}$ [panel a)] and $\chi_{|A\rangle}$ [panel b)] as ellipsoids in Stokes space. The coloured spheres give the measured atomic states for different input polarizations. The anticipated rotation of $R_x(\pi)$ for $\chi_{|A\rangle}$ is clearly visible when comparing both ellipsoids.

$$\chi_{|A\rangle} = \begin{pmatrix} 0.053 & 0.032 + 0.058i & 0.007 - 0.068i & -0.028 - 0.005i \\ 0.032 - 0.058i & 0.820 & 0.004 + 0.028i & -0.086 + 0.007i \\ 0.007 + 0.068i & 0.004 - 0.028i & 0.105 & -0.006 - 0.032i \\ -0.028 + 0.005i & -0.086 - 0.007i & -0.006 + 0.032i & 0.022 \end{pmatrix} \quad (5.3)$$

For this process a $R_x(\pi)$ rotation is expected, which would ideally result in $\chi_{11} = 1$. We find the largest entry of $\chi_{|A\rangle}$ to be $\chi_{11} = 0.82$ which is close to the expected rotation. This matrix entry directly relates to the process fidelity $F_{proc} = 82.0\%$ (see Section 2.2.2). Inserting F_{proc} into equation (2.10) yields an average state fidelity of $\bar{\mathcal{F}}_{|A\rangle} = 88.0\%$, which is in good agreement with the measured average fidelity and proves the quality of the maximum likelihood fit.

Experimental imperfections cause the observed deviation from unity fidelity. In the following, these imperfections and their expected impact on the measured fidelity are discussed in detail. Depending on the type of imperfection j , and the input polarization state, the atomic state is projected into an undesired state $|\psi_j\rangle$, which has a fidelity F_j with the desired state. Each of the experimental flaws has a probability p_j to occur. For this esti-

mation, the probabilities p_j are assumed to be small and we therefore neglect all correlated terms in the fidelity computation (see equation (2.7)) , but consider the imperfections independently. The effective reduction in fidelity is then calculated as $p_j(1 - F_j)$. The results of this discussion are summarized in Table 5.2. Another helpful resource which conveys a feeling for the expected impact is Figure 5.5, since one can simply compare the ideal expected state and the state which is caused by the imperfection.

Herald in $ D\rangle$	$ D\rangle$	$ A\rangle$	$ H\rangle$	$ V\rangle$	$ L\rangle$	$ R\rangle$	avg. impact
Mode matching	11 %	0 %	5.5 %	5.5 %	0 %	11 %	5.5 %
Atomic state manipulation	1.3 %	0 %	3.2 %	3.2 %	3.2 %	3.2 %	2.3 %
Dark counts	1.2 %	1.2 %	1.2 %	1.2 %	0 %	2.3 %	1.2 %
PBS extinction ratio	1 %	1 %	1 %	1 %	0 %	0 %	0.67 %
Two photon events	0 %	0 %	1.9 %	1.9 %	0 %	3.7 %	1.2 %
Birefringence	0 %	0 %	3 %	3 %	3 %	3 %	2 %
Sum of all errors	14.5 %	2 %	16 %	16 %	6 %	23 %	13 %

Herald in $ A\rangle$	$ D\rangle$	$ A\rangle$	$ H\rangle$	$ V\rangle$	$ L\rangle$	$ R\rangle$	avg. impact
Mode matching	0 %	11 %	5.5 %	5.5 %	0 %	11 %	5.5 %
Atomic state manipulation	0 %	1.3 %	3.2 %	3.2 %	3.2 %	3.2 %	2.3 %
Dark counts	1.2 %	1.2 %	1.2 %	1.2 %	0 %	2.3 %	1.2 %
PBS extinction ratio	1 %	1 %	1 %	1 %	0 %	0 %	0.67 %
Two photon events	0 %	0 %	1.9 %	1.9 %	0 %	3.7 %	1.2 %
Birefringence	0 %	0 %	0 %	0 %	0 %	0 %	0 %
Sum of all errors	2 %	14.5 %	13 %	13 %	3 %	20 %	11 %

Table 5.2.: Summary of experimental imperfections and their expected reduction of measured state fidelity. The highlighted percentage in the lower right corner denotes the expected reduction in average state fidelity.

Geometric mode matching ($p_{mm} = 11(3) \%$): The major contribution to experimental imperfections stems from the geometric mismatch of the impinging photons and the Gaussian cavity mode. By determining the coupling efficiency of light, which is transmitted through the resonator, into an optical fibre, the unmatched fraction of photons is measured to be $\eta = 8(3) \%$. Assuming a reflectivity of $\eta_{\text{ref}} = 69 \%$ for the matched fraction, gives a probability for detecting an unmatched photon $p_{mm} = \frac{\eta}{\eta + (1 - \eta) \cdot \eta_{\text{ref}}} = 11(3) \%$. These photons will not acquire a phase shift and therefore leave the atom in its initial state $|\uparrow_x\rangle$. Depending on the expected atomic state after reflection and detection, F_j varies from 1 for $|L\rangle$ polarized photons to 0 for right circular polarization. This imperfection does not apply to events with in-

herent postselection, i.e. events where input photon and herald are orthogonal with respect to each other, as the polarization flip from $|D\rangle$ to $|A\rangle$ already implies that the photon coupled to the cavity mode. The opposite case where herald photon and input photon are in the same state is thus affected twofold. Therefore, the reduction of the average state fidelity is $p_{mm} \cdot (2 \cdot 1 + 2 \cdot 0.5 + 2 \cdot 0)/6 = 5.5(2) \%$.

Atomic state preparation, rotation and read-out: The quality of atomic state manipulation can be separated into two parts: The preparation (**I**) and the rotation (**II**) of the atomic state. The impact of both imperfections is deducted from the Ramsey spectroscopy in Figure 4.8a.

I: The Ramsey fringes are offset to lower transfer probabilities by $p_{\text{prep}} = 1.3(0.3) \%$. We deduce that with this probability, the atom is prepared in an uncoupled Zeeman level of the $F = 2$ hyperfine-manifold that is not addressed by the Raman beams, most likely $|2, 1\rangle$. In this case, the atomic transition is detuned from the resonator frequency and any input photon enters and leaves the resonator without an interaction, independent of the photonic polarization state. Fluorescence hyperfine-state detection yields the atomic state $|\uparrow\rangle$. Consider an event where herald and input photon were in the same polarization state. Ideally, one expects to measure the atom in $|\downarrow\rangle$. Therefore, the reduction in fidelity is unity for this case. Now consider the case when a polarization flip is measured, e.g. $|D \uparrow\rangle \mapsto |A \uparrow\rangle$. Here, the expected atomic state corresponds to the measured state and the reduction in fidelity is zero. The overlap of the detected state $|\uparrow\rangle$ with every other state is 0.5, resulting in an overall reduction in average state fidelity of $p_{\text{prep}} \cdot (1 \cdot 1 + 0 \cdot 1 + 4 \cdot 0.5)/6 = 0.65(0.2) \%$

II: The quality of the atomic state rotation and the impact of dephasing is given by the visibility of the Ramsey fringes in Figure 4.8a. The atom is measured in the expected state in 95(1) % of the trials. The remaining $p_{\text{rot}} = 5(1) \%$ of all events result in a statistical mixture of $|\downarrow\rangle$ and $|\uparrow\rangle$. In the following, we assume that the first Raman $\frac{\pi}{2}$ -pulse prepared the atom in an imperfect state $\alpha|\downarrow\rangle + \beta|\uparrow\rangle$, which is close to the desired state $|\uparrow_x\rangle$. Reflection of a $|D\rangle$ -polarized input photon generates an entangled state: $\alpha|D \downarrow\rangle + \beta|A \uparrow\rangle$ (see also Section 5.5). The measurement of the photonic polarization state in D or A thus projects the atomic state into the desired state for storage and the reduction in fidelity equals zero. The same consideration is valid for $|A\rangle$ -polarized input photons. However, for every other input polarization one tries to measure a balanced atomic superposition state. Here, the error in the Raman pulse area and magnetic-field fluctuations lead to the measurement of a fully mixed state with the probability p_{rot} . Hence, the reduction in average state fidelity is $p_{\text{rot}} (4 \cdot 0.5)/6 = 1.7(0.3) \%$.

Dark counts ($p_{\text{dark}} = 2.3\%$): Evaluating background count rates results in a dark count probability of $p_{\text{dark}} = 2.3 \%$. Dark counts do not affect the atomic state and therefore result in the same detrimental reduction of fidelity as unmatched photons and leave

the atom in $|\uparrow_x\rangle$. These events correspond to a fully mixed state and give an average reduction in fidelity of $p_{\text{dark}} (4 \cdot 0.5 + 1 \cdot 0 + 1 \cdot 1) / 6 = 1.2\%$.

Beam splitter extinction ratio ($p_{bs} = 1\%$): The extinction ratio of the polarizing beam splitters in the detection setup was determined to be 1 : 100 in reflection and 1 : 1000 in transmission. Measuring a reflected photon in the wrong polarization basis has no effect on $|L\rangle$ and $|R\rangle$ -polarized photons due to the nature of the induced $R_x(\pi)$ rotation. In return, the reduction in fidelity for the four remaining input states is unity. This leads to a degradation in average state fidelity of $p_{bs} (4 \cdot 1 + 2 \cdot 0) / 6 = 0.67\%$.

Two photon events: The average impinging photon number of 0.082 can be inferred from the detected number of single photon events (0.03), the reflectivity of the setup η_{ref} and the efficiency of the detection setup (56%). From the Poissonian photon statistics of coherent pulses, we deduct that 4.9% of the detected single photon events are due to two photons in the incident laser pulse.

I: In about half of two-photon events, both photons are reflected ($p_{2ph2ref} = 2.5\%$). If the input polarization was linearly polarized, a GHZ-state [84] is generated. This type of entangled state exhibits classical correlations when tracing over a qubit subspace, e.g. by not detecting one of the photons, which would result in the following mixture for $|D\rangle$ -polarized input photons: $\rho = \frac{1}{2}(|D\downarrow\rangle\langle D\downarrow| + |A\uparrow\rangle\langle A\uparrow|)$. For input photons in $|D\rangle$ or $|A\rangle$, the generated classical state returns the expected correlations of the ideal state and the fidelity is unity. However, $|H\rangle$ and $|V\rangle$ result in the same type of mixture which shows a reduction in fidelity to 0.5 when compared to the expected state in the D/A polarization basis: $\rho_{|H\rangle} = \frac{1}{2}|D\downarrow_y + A\uparrow_y\rangle\langle D\downarrow_y + A\uparrow_y|$. For two reflected $|R\rangle$ -photons the induced phase shift is 2π leading to $F_u = 0$. The reflection of $|L\rangle$ -polarized photons does not have an impact on the atomic state, thus leading to unity fidelity in this case. This results in an overall reduction in fidelity due to two photon reflection of $p_{2ph2ref}(0.5 \cdot 2 + 1 \cdot 1 + 3 \cdot 0) / 6\% = 0.8\%$.

II: For the other half of the events, one of the two photons is absorbed in the atom-cavity system ($p_{2ph1abs} = 2.4\%$). While uncoupled polarization components do not affect the atomic state, coupled components fully project the atom into $|\uparrow\rangle$ or $|\downarrow\rangle$. Thus the reduction in fidelity for storing $|L\rangle$ polarized photons equals zero. Consider the case where two $|D\rangle$ -polarized input photons are impinging onto the atom-cavity system, which generates the state $\frac{1}{\sqrt{2}}(|DD\downarrow\rangle + |AA\uparrow\rangle)$. Projecting the atom into one of the two states $|\downarrow\rangle$ or $|\uparrow\rangle$ by absorbing one photon still yields the desired correlations between atomic state and photonic polarization. The same holds for an $|A\rangle$ -polarized input photon. This leads to no reduction in fidelity for the input states $|A\rangle$ and $|D\rangle$. Photon loss for every other polarization type returns a fully mixed state and the fidelity for these events is $F = 0.5$. Therefore, the average state fidelity is expected to be reduced by $p_{2ph1abs}(2 \cdot 0.5 \cdot 0.5 + 1 \cdot 0.5 + 3 \cdot 0) / 6\% = 0.4\%$.

Birefringence ($p_{bf} = 100\%$): Cavity birefringence generates a systematic rotation for her-

adds in $|D\rangle$ and a reduced fidelity of 3 % for the input states $|H\rangle, |V\rangle, |R\rangle$ and $|L\rangle$. The overall reduction in fidelity is $4 \cdot 3/6 \% = 2 \%$.

Adding these error sources up, leads to an overall fidelity reduction of 11(2) % (13(2) %) for heralding photons in $|A\rangle$ ($|D\rangle$), which is in agreement with the observed average state fidelity of 89(1) % (87(2) %).

5.4. Read-out: Atom-to-photon state mapping

In the previous section, the storage of a polarization qubit in the spin state of a single atom was demonstrated. In order to show the potential application of the system as an optical quantum memory, the reverse process, i.e. a transfer of the atomic onto a photonic state, is presented in this section. To this end, we follow the read-out protocol of Section 3.2. The experimental sequence is described in Section 5.4.1. In Section 5.4.2 the experimental results are presented and discussed.

5.4.1. Implementation of the read-out protocol

This section describes the experimental implementation of the read-out protocol. Figure 5.8 presents a schematic of the physical implementation and the temporal sequence of laser pulses.

The realization of the read-out protocol requires the use of previously introduced experimental techniques, which are described in detail in Figure 5.2. In contrast to the storage experiment, the polarization of the impinging photon remains $|D\rangle$ for all experimental runs and the atomic state is initialized in one of the six eigenstates of the Pauli spin operators. In this type of sequence, the atomic input state is randomly changed into one of these six states, when an atom is loaded. In order to perform state tomography on the polarization state of the photon, we change the basis of the detection setup during this period as well.

The atomic state detection, and thus the polarization of the output photon, has two possible outcomes. If the atom is measured in $|\downarrow\rangle$, the output polarization state corresponds to the desired state $|\varphi\rangle_p$. However, detecting the atom in $|\uparrow\rangle$, returns the rotated state $\sigma_3 |\varphi\rangle_p$. This conditional unitary transformation could be reversed by feedforward of the atomic state measurement outcome. To this end, the photon would have to be stored for $\sim 5 \mu\text{s}$, e.g. using a 1 km long optical fibre. In our experiment this feedforward polarization rotation is missing. Hence, the read-out process characterized here is not deterministic and the efficiency has an upper bound of 50 %.

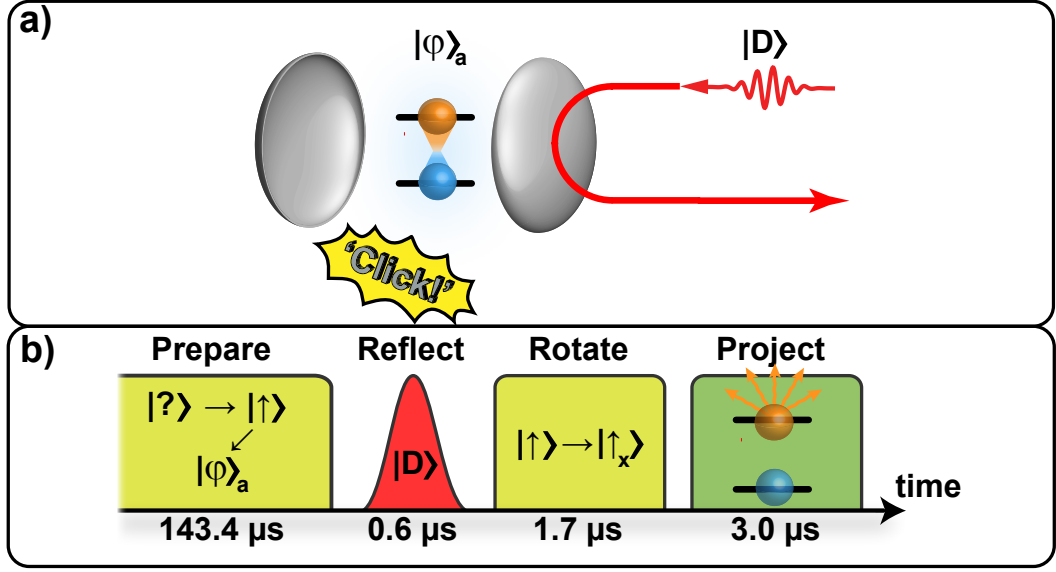


Figure 5.8.: **Implementation of the read-out protocol.** a) Physical implementation of the read-out protocol. A $|D\rangle$ -polarized photon is reflected off the atom-cavity-system with the atom being in an arbitrary state $|\varphi\rangle_a$. After reflection, the atomic state is detected in the $|\downarrow_x\rangle / |\uparrow_x\rangle$ basis and thus projects the entangled atom-photon state. Depending on the detected spin state the photonic state is either projected to $|\varphi\rangle_p$ or the rotated state $\sigma_3 |\varphi\rangle_p$. b) Temporal sequence of the individual steps needed to perform the read-out experiment. Yellow rectangles display atomic state manipulations, such as the initialization of the atom. The red Gaussian represents the reflection of a faint laser pulse with Gaussian envelope and the green rectangle expresses the projection of the atomic state via cavity enhanced hyperfine-state detection.

5.4.2. Characterization of the read-out process

After describing the experimental sequence, we turn to the evaluation of the read-out process. The efficiency of the read-out protocol is given by the reflectivity of the system on resonance and the fact that 50 % of the events return a rotated state. For the transfer of the atomic state onto a photon's polarization, we always reflect a $|D\rangle$ -polarized photon off the atom-cavity system. Taking the average reflectivity of the system of 69 % for $|D\rangle$ -polarized photons into account (see Figure 5.3), results in an overall read-out efficiency of 35(1) %.

In the storage experiment, the detection of a photon heralded a successful event. The opposite statement, i.e. that detection of the atomic state can serve as a herald for the successful reflection of a photon, is only true in specific cases with a known initial atomic state [36]. In a read-out scenario, the atom is considered to be in an arbitrary superpo-

sition state $|\varphi\rangle_a$. The unknown state population and phase relation does not allow for a combination of unitary rotations, which would deterministically indicate the successful reflection of a $|D\rangle$ -polarized photon. Therefore, the presented measurements have to be conditioned on the detection of a single photon after reflection of a weak coherent pulse.

Atomic input state	Atom in $ \uparrow\rangle$	Atom in $ \downarrow\rangle$	average
$ \downarrow_x\rangle$	76(2) %	95(1) %	85.5 %
$ \downarrow_y\rangle$	78(2) %	83(2) %	80.5 %
$ \downarrow\rangle$	95(2) %	98(1) %	96.5 %
$ \uparrow_x\rangle$	90(2) %	80(2) %	85 %
$ \uparrow_y\rangle$	85(2) %	85(2) %	85 %
$ \uparrow\rangle$	84(2) %	87(2) %	85.5 %
avg. fidelity $\overline{\mathcal{F}}$	$\overline{\mathcal{F}}_{ \uparrow\rangle} = \mathbf{85(1)} \%$	$\overline{\mathcal{F}}_{ \downarrow\rangle} = \mathbf{88(1)} \%$	86 %

Table 5.3.: Measured state fidelities of reconstructed polarization states with the ideal expected state for atomic state measurement outcomes $|\uparrow\rangle$ and $|\downarrow\rangle$. The given errors are derived via the Monte-Carlo technique (see Appendix B).

The process is characterized individually for the measurement outcomes of the atomic state $|\downarrow\rangle$ and $|\uparrow\rangle$. Essentially, a characterization scheme in analogy to the one described in Figure 5.3 is utilized to determine output states and state fidelities by quantum state tomography. The state fidelities with the expected ideal polarization state are computed using equation (2.7) and shown in Table 5.3. The reconstructed output states are used for quantum process tomography via the maximum likelihood algorithm (see Appendix A).

In Figure 5.9 a and b, the reconstructed processes, $\chi_{|\downarrow\rangle}$ and $\chi_{|\uparrow\rangle}$, are presented as deformed Poincaré spheres.

First, the process $\chi_{|\downarrow\rangle}$ is examined. Here, the reconstructed output polarizations give an average state fidelity of $\overline{\mathcal{F}}_{|\downarrow\rangle} = 88(1) \%$. The maximum likelihood fit algorithm is configured to return a trace-preserving process matrix. The process fidelity with the identity operation is $\chi_{00} = F_{proc} = 0.819$, which leads to an expected average state fidelity of 88 % (see equation (2.10)).

Upon detection of the atom in $|\uparrow\rangle$, we find a trace-preserving process matrix with the dominant entry $\chi_{33} = 0.77$, corresponding to the expected rotation of the polarization qubit by $R_z(\pi)$. The process fidelity of $\chi_{|\uparrow\rangle}$ is $F_{proc} = 0.773$ and results in a predicted average state fidelity $\overline{\mathcal{F}}_{|\uparrow\rangle} = 85 \%$. The measured and predicted average state fidelities are in good agreement. Both processes show the predicted rotation due to birefringence. However, we find the maximum fidelity for a slightly rotating process with a rotation angle of $|\xi| = 0.09\pi$, which is lower than the expected rotation of 0.11π . In contrast to the storage scheme, where an automatic phase reference is given by measuring heralds in

$|D\rangle$ and $|A\rangle$, the read-out scheme does not provide this reference since no $|A\rangle$ -polarized photons impinge onto the resonator. The reduction in rotation might thus originate from a slight undetected cavity frequency offset, which would rotate the read-out process for $|A\rangle$ -polarized photons by $|\xi| = 0.02\pi$. Two adjustments could be made to improve this result. First, one could stabilize the cavity frequency closer to the $|D\rangle$ -polarized eigenmode and thus mitigate the rotation. Second, a qubit redefinition for the storage scheme, i.e. rotating the atomic state on the detection of a $|D\rangle$ -polarized herald, would allow for a read-out with $|A\rangle$ -polarized photons.

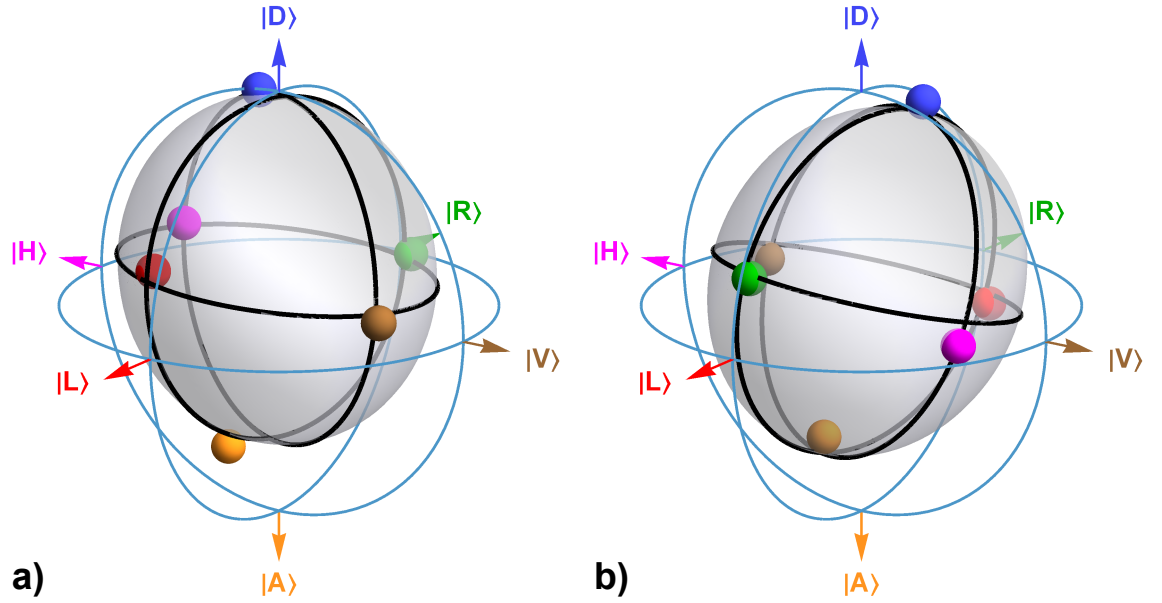


Figure 5.9.: **Visualization of both read-out processes, $\chi_{|\uparrow\rangle}$ and $\chi_{|\downarrow\rangle}$, as ellipsoids in Stokes space.** The coloured labels correspond to the position of the ideal output states for a direct mapping process. The coloured spheres show the position of the measured output states. Reconstruction of the process $\chi_{|\downarrow\rangle}$ [$\chi_{|\uparrow\rangle}$] is shown in **a)** [**b)**]. The rotation of the polarization qubit by $R_z(\pi)$ is clearly visible.

Experimental imperfections were discussed in Figure 5.3. The discrepancy between average state fidelities can be explained by inherent post selection on correct atomic state manipulation, as detecting the atom in $|\downarrow\rangle$ implies that atomic state preparation and rotation worked as intended. Table 5.4 presents a breakdown of the expected fidelity reduction for both process matrices. Taking all imperfections into account leads to an expected average state fidelity of 86(2) % (89(2) %), for detection of the atom in $|\uparrow\rangle$ ($|\downarrow\rangle$), which is within in the errors of the observed fidelities.

In summary, the transfer of an atomic qubit onto the polarization state of a photon was characterized. In principle, this read-out scheme is deterministic when feedforward is

applied to the photonic state. The classical boundary for a read-out process with a mean impinging photon number of $\bar{n} = 0.08$ and an efficiency of 35(1)% is 67.6%. Thus the observed state fidelities, which are post selected on the detection of a reflected photon, are clearly non-classical.

Atom in $ \uparrow\rangle$	$ \uparrow\rangle$	$ \downarrow\rangle$	$ \uparrow_x\rangle$	$ \downarrow_x\rangle$	$ \uparrow_y\rangle$	$ \downarrow_y\rangle$	avg. impact
Mode matching	11 %	0 %	0%	11 %	5.5 %	5.5 %	5.5 %
Atomic state manipulation	1.3 %	2.5 %	3.2 %	3.2 %	3.2 %	3.2 %	2.8 %
Dark counts	1.2 %	1.2 %	1.2 %	1.2 %	1.2 %	1.2 %	1.2 %
PBS extinction ratio	1 %	1 %	1 %	1 %	1 %	1 %	1 %
Two photon events	0 %	0 %	1.9 %	1.9 %	1.9 %	1.9 %	1.3 %
Birefringence	0 %	0 %	3 %	3 %	3 %	3 %	2 %
Sum of errors	14.5 %	5 %	10 %	21 %	16 %	16 %	14 %

Atom in $ \downarrow\rangle$	$ \uparrow\rangle$	$ \downarrow\rangle$	$ \uparrow_x\rangle$	$ \downarrow_x\rangle$	$ \uparrow_y\rangle$	$ \downarrow_y\rangle$	avg. impact
Mode matching	11 %	0 %	11%	0 %	5.5 %	5.5 %	5.5 %
Atomic state manipulation	0 %	0 %	0 %	0 %	0 %	0 %	0 %
Dark counts	1.2 %	1.2 %	1.2 %	1.2 %	1.2 %	1.2 %	1.2 %
PBS extinction ratio	1 %	1 %	1 %	1 %	1 %	1 %	1 %
Two photon events	0 %	0 %	1.9 %	1.9 %	1.9 %	1.9 %	1.3 %
Birefringence	0 %	0 %	3 %	3 %	3 %	3 %	2 %
Sum of errors	13 %	2 %	18 %	7 %	12.6 %	12.6 %	11 %

Table 5.4.: Summary of experimental imperfections and their expected reduction of measured state fidelity in the read-out process. The highlighted percentage in the lower right corner denotes the expected reduction in average state fidelity.

5.5. Generation of atom-photon entanglement

The results of this section will also be published in [37]. After demonstrating the write and read-out process of an optical quantum memory, an example of the versatility of the interaction mechanism is provided by the generation of atom-photon entanglement. This section employs the previously introduced experimental sequence in order to generate the maximally-entangled Bell state $|\Phi^+\rangle$. To this end, the atom is prepared in the state $|\uparrow_x\rangle$ and a linearly $|D\rangle$ -polarized photon is reflected off the atom-cavity system:

$$|D\rangle |\uparrow_x\rangle = \frac{1}{\sqrt{2}} |D\rangle (|\uparrow\rangle + |\downarrow\rangle) \rightarrow \frac{1}{\sqrt{2}} (|D \downarrow\rangle + |A \uparrow\rangle) = |\Phi^+\rangle \quad (5.4)$$

In order to perform quantum state tomography on the atom-photon entangled state the read-out sequence described in Section 5.4.1 has to be adjusted. Instead of measuring the atom only in the $|\uparrow_x\rangle/|\downarrow_x\rangle$ basis, bases along three orthogonal directions need to be taken into account and the common set of Raman rotations ($R_y(\frac{\pi}{2})$, $R_{-x}(\frac{\pi}{2})$, $R_x(0)$) is employed to perform tomography on the atomic state. Due to a slightly different optical pumping power during this experiment, we find new filtering parameters for the evaluation of the data. These are expressed in the notation of equation (5.1): $\{\zeta_{multi}, \zeta_{loss}, \zeta_{|\uparrow\rangle}\} = \{15, 8, 3\}$.

Due to the fact that $|\Phi^+\rangle$ is generated from faint laser pulses which have a low average photon number, events are only evaluated if a photon was detected in the expected temporal region of interest. We perform two-particle quantum state tomography using a maximum likelihood algorithm [49] and reconstruct the combined atom-photon state. The real and imaginary part of the reconstructed density matrix are shown in Figure 5.10a and b respectively.

The fidelity with the ideal state is 81(1) %, where the standard error has been determined via the Monte-Carlo technique (see Appendix B). This value clearly surpasses the classical boundary of 50 %, which is derived from the fidelity of the ideal state with a classically correlated state, e.g. $|A \uparrow\rangle$. Furthermore the phase of the reconstructed state is slightly rotated. We find a maximum fidelity of 83 % for a state of the form $(e^{-i\xi} |D \downarrow\rangle + |A \uparrow\rangle)/\sqrt{2}$, with an angle of $\xi = 0.11 \pi$, which is in agreement with the rotation of 0.11π expected from cavity birefringence (see Section 3.3.3).

In the spirit of the previous discussion of imperfections in Section 5.3, the influence of experimental flaws on the measured state fidelity is estimated in the following paragraph [37].

Geometric mode matching ($p_{mm} = 11(3) \%$): The unmatched fraction of photons will not acquire a phase shift and therefore result in a separable state $|D\rangle (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$, which has a fidelity with the ideal state of 0.25. Therefore the reduction in fidelity is $p_{mm}(1 - 0.25) = 8(3) \%$.

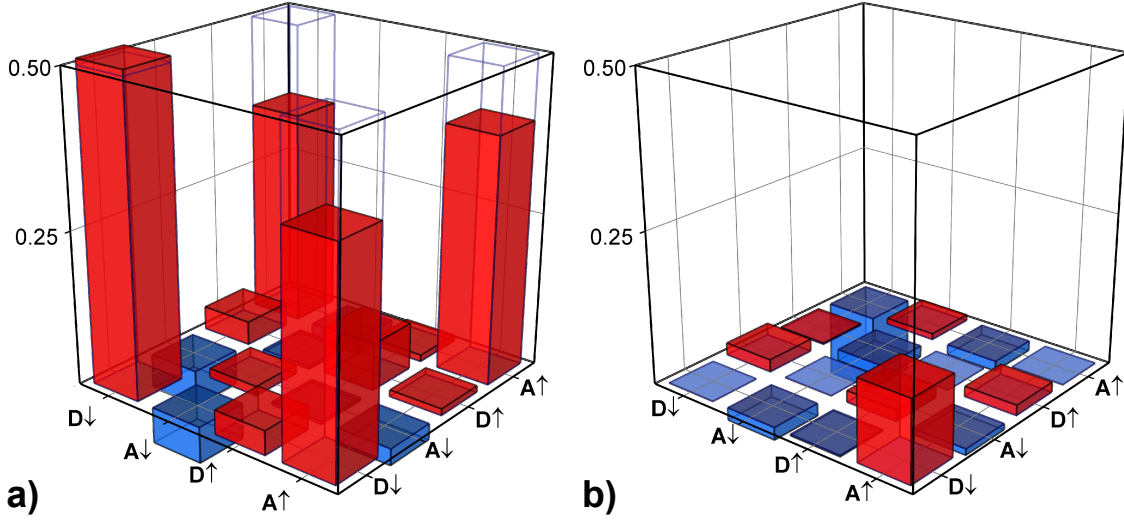


Figure 5.10.: **Reconstructed density matrix ρ .** **a)** Real part of ρ . To increase visibility, the data is colour-coded. Red bars indicate positive values, while blue bars correspond to values equal to or smaller than zero. The blue wire-frames indicate the ideally expected Bell state $|\Phi^+\rangle$ **b)** Imaginary part of the density matrix. As expected, most of the elements are close to zero except for $|A \uparrow\rangle\langle D \downarrow|$ and $|D \downarrow\rangle\langle A \uparrow|$. These values indicate a phase rotation of 0.11π which can be explained by resonator birefringence.

Atomic state preparation, rotation and read out ($p_{atom} = 5(1)\%$): The quality of atomic state manipulation leads to a reduction in fidelity of $5(1)\%$, as the atom is measured in an orthogonal state.

Dark counts ($p_{dark} = 3\%$): These events correspond to a fully mixed state, which has a fidelity with the expected state of 0.25 and thus leads to a fidelity reduction of $p_{dark}(1 - 0.25) = 2\%$.

Beam splitter extinction ratio ($p_{bs} = 1\%$): Detecting a photon in the wrong polarization basis yields an orthogonal state, leading to a reduction in fidelity of 1% .

Two photon events ($p_{2ph} = 4\%$): In about half of the cases two photons are reflected. Not detecting one of the two photons results in a mixture with classical correlations which has a fidelity with the expected state of 0.5. Resulting in a reduction in fidelity of $(1 - 0.5) \cdot 2\% = 1\%$.

For the other half of the events, one of the two photons is absorbed in the atom-cavity system. While uncoupled polarization components do not affect the atomic state, coupled components fully project the atom into $|\uparrow\rangle$ or $|\downarrow\rangle$, which leads to the pure states $|A \uparrow\rangle$ or $|D \downarrow\rangle$. Both states have a fidelity with the ideal entangled state of 0.5. Resulting in another reduction in fidelity of 1% .

Summing over the impact of all imperfections leads to an expected fidelity of 83 %, which is in good agreement measured value taking the rotation of the state into account.

6. Summary and outlook

In this thesis, quantum state mapping between a flying optical photon and a single atom trapped at the centre of an optical cavity has been demonstrated. The employed scheme is based on a controlled-phase gate between the polarization of the photon and the spin of the atom. The demonstrated mapping of the photonic state onto the atomic state corresponds to the *heralded storage* of a flying qubit in a stationary qubit. With the implementation of feedforward onto the atomic state, the storage-scheme is conceptually deterministic. In addition, the transfer of a well-prepared atomic state onto the polarization of a photon was demonstrated, corresponding to the *read-out* of a stationary qubit.

The quantum-mechanical nature of the storage and the read-out process has been proven by average state fidelities well beyond the classical boundary. Combining these two processes would result in an optical quantum memory with intrinsically heralded storage. In the following, experimental techniques are highlighted which would foremost improve the performance of the demonstrated processes and additionally pave the way for the study of further quantum phenomena in our system.

Two experimental imperfections had a major impact on the measured fidelities. First, the geometric mode-matching of the impinging photons and the resonator mode led to a detrimental fraction of uncoupled photons. This resulted in an even stronger decrease in fidelity when reflecting two photons off the atom-cavity system [37]. A future task is therefore the high-precision alignment and profile-shaping of the impinging beam such that it matches the Gaussian resonator mode. Second, imperfect preparation of the atom in the coupled state reduced the control over the system. One might overcome this obstacle by the employment of fibre Fabry-Perot resonators which should allow for an increased state-discrimination fidelity since they potentially provide a larger coupling strength and therefore result in an improved control over the atomic state [85]. The resulting increase in fidelity would allow for the realization of experiments which are based on the reflection of more than two photons, i.e. the generation of photonic matrix product states [86].

Heralding a successful storage event is only accomplished if the photon is reflected off the atom-cavity system and detected afterwards. Both parameters, the reflectivity and the detection efficiency, could be improved. A higher reflectivity of the system could be obtained by combining a larger coupling strength g with a transmissivity of the outcoupling mirror that is more dominant compared to all other cavity loss mechanisms. This goal might be accomplished using the aforementioned fibre resonators [87, 88]. The detection

efficiency of the photonic herald could also be improved by a factor of 1.7 if high-efficiency superconducting single-photon detectors [89] were used in the experiment. This could also provide an extension of the previously demonstrated nondestructive photon detection using our setup [36]: In this experiment, an odd-even photon-number measurement was performed on impinging pulses with a low average photon number. For larger impinging photon numbers, the use of highly efficient detectors in the setup could underpin the number parity predicted by the atom-cavity detector with an independent and precise measurement. In principle, the odd-even parity measurement could be extended to the preparation of optical Fock states. In analogy to experiments with trapped microwave photons in superconducting cavities and Rydberg atoms traversing the cavity [90, 91], a distribution of phase shifts per photon for several concatenated detectors would be required for this experiment. Assuming negligible loss, this could be achieved by slightly detuning the atom-cavity system from the frequency of the impinging pulse.

The efficiency of our implementation of the read-out protocol could be increased twofold by a polarization rotation which is conditioned on the outcome of the atomic state detection. As the detection of the atomic state takes a finite amount of time, the photon would need to be stored in an optical fibre after reflection off the atom-cavity system. Once the atomic state is detected, the photonic polarization state could be easily rotated with an electro-optic modulator which is positioned after the fibre. Implementing this type of feedforward would also pave the way for the realization of a deterministic photon-photon controlled-phase gate utilizing a quantum eraser scheme [92, 93].

An obvious application of the presented heralded memory is a heralded quantum-state transfer between nodes of an elementary quantum network [29]. Towards this end, two distant atom-cavity systems would have to be linked via an optical fibre. A photon is reflected off the sender node and coupled into the fibre link. Reflection of the photon off the second atom-cavity system, i.e. the receiver, and subsequent detection of the photon would herald a successful quantum-state transfer. We point out that this experiment would not require an optical delay line. One would just rotate the state of the receiver node conditioned on both measurement outcomes, i.e. the photonic polarization and the state of the atom at the sender node. Moreover, the heralded distribution of entanglement could be realized by preparing both atoms in a superposition state and reflecting a photon off both nodes. Conceptually, such a heralded distribution of entanglement would be an ideal component of a quantum repeater architecture [22]. Furthermore, the reflection of a photon off several atom-cavity systems allows for the generation of atomic cluster states [94] that might be used for entanglement-enhanced precision measurements [95]. Apart from progress in the field of quantum information processing, heralded distribution of entanglement could also provide a way for performing loophole-free Bell tests [96, 97, 98, 99].

A. Maximum likelihood technique

Statistical errors and measurement uncertainties might result in a reconstructed quantum process which is not a completely positive map and therefore not physical. A standard technique to numerically determine a physical process is the maximum likelihood technique. Here one tries to find a physical process which replicates the measured data as good as possible. This type of estimation has also been applied in [43]. Other resources for the following discussion are [49, 100].

For the maximum likelihood estimation it is convenient to switch to the matrix representation and describe the process with the process matrix and parametrize it to be Hermitian $\chi = T^\dagger T$. The Hermitian parametrization guarantees a positive map. A starting point for the algorithm is to analytically calculate (see e.g. [5]) the process matrix and then perform a LU decomposition of the analytical result in order to obtain the matrix T in the lower tri-diagonal form.

The goal of the algorithm is then to find a process matrix χ which predicts the measured detection events N by giving a expected number of events N_χ that is close to N . To this end, one has to take all detection events for all input states (ρ_{in}), detection bases (b) and both measurement outcomes ($\pm$) into account. The measured events are denoted as $N[\rho_{in}, b, \pm]$ and the total measured events in a single basis are denoted as $N_b[\rho_{in}] = N[\rho_{in}, b, +] + N[\rho_{in}, b, -]$. The expected output density matrix of the process ρ_{out} is derived with equation (2.3). One is then able to determine the expected counts from the process by equation (2.5): $N_\chi(\rho_{in}, b, \pm) = N_b(\rho_{in}) \cdot \langle b, \pm | \rho_{out} | b, \pm \rangle$. The probability to measure a certain number of counts is assumed to be a Gaussian distribution $p \propto \exp[-(N - N_\chi)^2/\sigma^2]$. The standard deviation is derived from the Poisson distribution $\sigma = \sqrt{N+1} \approx \sqrt{N}$. The overall probability for the reconstructed process to correspond to the measured process is then proportional to the product of every single detection probability

$$p(\chi) \propto \exp \left[- \sum_{\rho_{in}, b, \pm} \frac{(N[\rho_{in}, b, \pm] - N_\chi[\rho_{in}, b, \pm])^2}{N[\rho_{in}, b, \pm]} \right],$$

which has to be maximized. Since the exponential function is monotonous, this is equivalent to the maximization of the exponent.

So far, a process is evaluated which does not necessarily preserve the trace but only fits the data as good as possible. We implement an additional boundary condition which has

to be minimized in order to obtain a trace-preserving process [101]

$$p_{Tr} = \sum_{i,j,k=0}^3 \left| \chi_{i,j} \text{Tr} [\sigma_j^\dagger \sigma_i \sigma_k] - \delta_{k,0} \right|.$$

$\delta_{k,0}$ is the Kronecker delta. Therefore the algorithm needs to find a minimum for the combined expression

$$\sum_{\rho_{in}, b, \pm} \frac{(N[\rho_{in}, b, \pm] - N_\chi[\rho_{in}, b, \pm])^2}{N[\rho_{in}, b, \pm]} + \alpha_{Tr} p_{Tr}.$$

Here, α_{Tr} denotes a weighting factor that either puts emphasis on the fitting of a trace-preserving process ($\alpha_{Tr} \gg 0$) within the precision of the fit algorithm or approximates the measured states without any additional boundary condition ($\alpha_{Tr} = 0$). In the presented work, the obtained process matrices were obtained by incrementally increasing α_{Tr} until the algorithm returned a process which had the highest overlap with the measured data and was found to be trace-preserving. The approximation to the measured data is verified by calculating the fidelity of the expected states from the fitted process and the measured states.

B. Monte Carlo method

The Monte Carlo method is a numerical technique which has broad applications in the analysis of data and simulation [102]. A crude analogy to this approach is to “gamble” for further experimental results which are statistically spread around the measured values. In the presented work, the Monte Carlo method was used to numerically approximate the standard deviation of the measured state fidelities. Determination of the error on the measured fidelity for an input state ρ_{in} is accomplished by implementing the algorithm which is comprised of four steps:

1. Define a starting point by recording the photonic or atomic detection events N and calculate the fidelity $F[\rho_{exp}[N], \rho_{in}]$ with the desired state (see equation (2.7)).
2. Generate a new set of data by distributing random Poissonian noise over the measured data. The detection of photonic and atomic states is discrete, meaning that one measures the atomic state either in $|\downarrow\rangle$ or $|\uparrow\rangle$ and the photon in one of two polarizations. Therefore the standard deviation around the mean value of N detected events is $\sigma = \sqrt{N+1} \approx \sqrt{N}$. The distribution of statistical fluctuations over the measured data results in a new set of events which is denoted as N_i .
3. Calculate the fidelity for the new set of data N_i with $F_i = F[\rho_{exp}[N_i], \rho_{in}]$.
4. Perform steps two and three m times and accumulate the calculated fidelities of each iteration.

Then, by taking all accumulated fidelity values into account, one is able to approximate the standard deviation from the measured fidelity by calculating the square root of the variance of the randomized set of results.

$$\Delta F = \sqrt{\frac{\sum_{i=1}^m (F_i - F_0)^2}{N}}.$$

Bibliography

- [1] D. Hanneke, S. Fogwell and G. Gabrielse. New measurement of the electron magnetic moment and the fine structure constant. *Physical Review Letters* **100**, 120801 (2008).
- [2] W. Liu, M. G. Boshier, S. Dhawan, O. van Dyck, P. Egan, X. Fei, M. Grosse Perdekamp, V. W. Hughes, M. Janousch, K. Jungmann, D. Kawall, F. G. Mariam, C. Pillai, R. Prigl, G. zu Puttlitz, I. Reinhard, W. Schwarz, P. A. Thompson and K. A. Woodle. High precision measurements of the ground state hyperfine structure interval of muonium and of the muon magnetic moment. *Physical Review Letters* **82**, 711–714 (1999).
- [3] W. Shockley. A unipolar "field-effect" transistor. *Proceedings of the IRE* **40**, 1365–1376 (1952).
- [4] T. H. Maiman. Stimulated optical radiation in ruby. *Nature* **187**, 493–494 (1960).
- [5] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press (2010).
- [6] T. D. Ladd, F. Jelezko, R. Laflamme, Y. Nakamura, C. Monroe and J. L. O'Brien. Quantum computers. *Nature* **464**, 45–53 (2010).
- [7] R. P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics* **21**, 467–488 (1982).
- [8] S. Lloyd. Universal quantum simulators. *Science* **273**, 1073–1078 (1996).
- [9] N. Gisin and R. Thew. Quantum communication. *Nature Photonics* **1**, 165–171 (2007).
- [10] C. H. Bennett and D. P. DiVincenzo. Quantum information and computation. *Nature* **404**, 247–255 (2000).
- [11] P. W. Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing* **26**, 1484–1509 (1997).
- [12] R. L. Rivest, A. Shamir and L. Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Commun. ACM* **21**, 120–126 (1978).
- [13] E. Lucero, R. Barends, Y. Chen, J. Kelly, M. Mariantoni, A. Megrant, P. O'Malley, D. Sank, A. Vainsencher, J. Wenner, T. White, Y. Yin, A. N. Cleland and J. M.

- Martinis. Computing prime factors with a Josephson phase qubit quantum processor. *Nature Physics* **8**, 719–723 (2012).
- [14] L. M. K. Vandersypen, M. Steffen, G. Breyta, C. S. Yannoni, M. H. Sherwood and I. L. Chuang. Experimental realization of Shor’s quantum factoring algorithm using nuclear magnetic resonance. *Nature* **414**, 883–887 (2001).
- [15] A. Politi, J. C. F. Matthews and J. L. O’Brien. Shor’s quantum factoring algorithm on a photonic chip. *Science* **325**, 1221–1221 (2009).
- [16] J. I. Cirac, A. K. Ekert, S. F. Huelga and C. Macchiavello. Distributed quantum computation over noisy channels. *Physical Review A* **59**, 4249–4254 (1999).
- [17] H. J. Kimble. The quantum internet. *Nature* **453**, 1023–1030 (2008).
- [18] A. I. Lvovsky, B. C. Sanders and W. Tittel. Optical quantum memory. *Nat Photon* **3**, 706–714 (2009).
- [19] C. Simon, M. Afzelius, J. Appel, A. B. d. l. Giroday, S. J. Dewhurst, N. Gisin, C. Y. Hu, F. Jelezko, S. Kröll, J. H. Müller, J. Nunn, E. S. Polzik, J. G. Rarity, H. D. Riedmatten, W. Rosenfeld, A. J. Shields, N. Sköld, R. M. Stevenson, R. Thew, I. A. Walmsley, M. C. Weber, H. Weinfurter, J. Wrachtrup and R. J. Young. Quantum memories. *The European Physical Journal D* **58**, 1–22 (2010).
- [20] N. Gisin, G. Ribordy, W. Tittel and H. Zbinden. Quantum cryptography. *Reviews of Modern Physics* **74**, 145–195 (2002).
- [21] G. Brassard, N. Lütkenhaus, T. Mor and B. C. Sanders. Limitations on practical quantum cryptography. *Physical Review Letters* **85**, 1330–1333 (2000).
- [22] H.-J. Briegel, W. Dür, J. I. Cirac and P. Zoller. Quantum repeaters: The role of imperfect local operations in quantum communication. *Physical Review Letters* **81**, 5932–5935 (1998).
- [23] N. Sangouard, C. Simon, H. de Riedmatten and N. Gisin. Quantum repeaters based on atomic ensembles and linear optics. *Reviews of Modern Physics* **83**, 33–80 (2011).
- [24] T. Chanelière, D. N. Matsukevich, S. D. Jenkins, S.-Y. Lan, T. A. B. Kennedy and A. Kuzmich. Storage and retrieval of single photons transmitted between remote quantum memories. *Nature* **438**, 833–836 (2005).
- [25] X.-H. Bao, A. Reingruber, P. Dietrich, J. Rui, A. Dück, T. Strassel, L. Li, N.-L. Liu, B. Zhao and J.-W. Pan. Efficient and long-lived quantum memory with cold atoms inside a ring cavity. *Nature Physics* **8**, 517–521 (2012).
- [26] M. Afzelius, I. Usmani, A. Amari, B. Lauritzen, A. Walther, C. Simon, N. Sangouard, J. Minář, H. de Riedmatten, N. Gisin and S. Kröll. Demonstration of atomic frequency comb memory for light with spin-wave storage. *Physical Review Letters* **104**, 040503 (2010).

- [27] A. D. Boozer, A. Boca, R. Miller, T. E. Northup and H. J. Kimble. Reversible state transfer between light and a single trapped atom. *Physical Review Letters* **98**, 193601 (2007).
- [28] H. P. Specht, C. Nölleke, A. Reiserer, M. Uphoff, E. Figueroa, S. Ritter and G. Rempe. A single-atom quantum memory. *Nature* **473**, 190–193 (2011).
- [29] S. Ritter, C. Nölleke, C. Hahn, A. Reiserer, A. Neuzner, M. Uphoff, M. Mücke, E. Figueroa, J. Bochmann and G. Rempe. An elementary quantum network of single atoms in optical cavities. *Nature* **484**, 195–200 (2012).
- [30] L.-M. Duan, M. D. Lukin, J. I. Cirac and P. Zoller. Long-distance quantum communication with atomic ensembles and linear optics. *Nature* **414**, 413–418 (2001).
- [31] L.-M. Duan and C. Monroe. Colloquium: Quantum networks with trapped ions. *Reviews of Modern Physics* **82**, 1209–1224 (2010).
- [32] D. N. Matsukevich, T. Chanelière, S. D. Jenkins, S.-Y. Lan, T. A. B. Kennedy and A. Kuzmich. Entanglement of remote atomic qubits. *Physical Review Letters* **96**, 030405 (2006).
- [33] C.-W. Chou, J. Laurat, H. Deng, K. S. Choi, H. d. Riedmatten, D. Felinto and H. J. Kimble. Functional quantum nodes for entanglement distribution over scalable quantum networks. *Science* **316**, 1316–1320 (2007).
- [34] J. I. Cirac, P. Zoller, H. J. Kimble and H. Mabuchi. Quantum state transfer and entanglement distribution among distant nodes in a quantum network. *Physical Review Letters* **78**, 3221–3224 (1997).
- [35] L.-M. Duan and H. J. Kimble. Scalable photonic quantum computation through cavity-assisted interactions. *Physical Review Letters* **92**, 127902 (2004).
- [36] A. Reiserer, S. Ritter and G. Rempe. Nondestructive detection of an optical photon. *Science* **342**, 1349–1351 (2013).
- [37] A. Reiserer, N. Kalb, G. Rempe and S. Ritter. A quantum gate between flying optical photons and a single trapped atom. *Nature (in press)*, (2014).
- [38] Reiserer, Andreas. *A controlled phase gate between a single atom and an optical photon*. Dissertation, Technische Universität München (2014).
- [39] L. Isenhower, E. Urban, X. L. Zhang, A. T. Gill, T. Henage, T. A. Johnson, T. G. Walker and M. Saffman. Demonstration of a neutral atom controlled-NOT quantum gate. *Physical Review Letters* **104**, 010503 (2010).
- [40] D. C. Burnham and D. L. Weinberg. Observation of simultaneity in parametric production of optical photon pairs. *Physical Review Letters* **25**, 84–87 (1970).
- [41] A. Ourjoumtsev, R. Tualle-Brouiri, J. Laurat and P. Grangier. Generating optical schrödinger kittens for quantum information processing. *Science* **312**, 83–86 (2006).

- [42] A. Ourjoumtsev, H. Jeong, R. Tualle-Brouri and P. Grangier. Generation of optical ‘Schrödinger cats’ from photon number states. *Nature* **448**, 784–786 (2007).
- [43] H. Specht. *Einzelatom-Quantenspeicher für Polarisations-Qubits*. Dissertation, Technische Universität München (2010).
- [44] H. Tanji, S. Ghosh, J. Simon, B. Bloom and V. Vuletić. Heralded single-magnon quantum memory for photon polarization states. *Physical Review Letters* **103**, 043601 (2009).
- [45] C. Kurz, M. Schug, P. Eich, J. Huwer, P. Müller and J. Eschner. High-fidelity heralded photon-to-atom quantum state transfer. *arXiv:1312.5995 [quant-ph]*, (2013).
- [46] M.-D. Choi. Completely positive linear maps on complex matrices. *Linear Algebra and its Applications* **10**, 285–290 (1975).
- [47] U. Fano. A Stokes-parameter technique for the treatment of polarization in quantum mechanics. *Physical Review* **93**, 121–123 (1954).
- [48] S. Gasiorowicz. *Quantum physics*. Wiley, Hoboken, N.J. (2003).
- [49] J. Altepeter, E. Jeffrey and P. Kwiat. Photonic State Tomography. In: P. R. Berman and C. C. Lin, editors, *Advances In Atomic, Molecular, and Optical Physics*, Vol. Volume 52, 105–159. Academic Press, (2005).
- [50] A. Uhlmann. The “transition probability” in the state space of a $*$ -algebra. *Reports on Mathematical Physics* **9**, 273–279 (1976).
- [51] R. Jozsa. Fidelity for mixed quantum states. *Journal of Modern Optics* **41**, 2315–2323 (1994).
- [52] M. D. Bowdrey, D. K. L. Oi, A. J. Short, K. Banaszek and J. A. Jones. Fidelity of single qubit maps. *Physics Letters A* **294**, 258–260 (2002).
- [53] A. Gilchrist, N. K. Langford and M. A. Nielsen. Distance measures to compare real and ideal quantum processes. *Physical Review A* **71**, 062310 (2005).
- [54] C. Nölleke. *Quantum state transfer between remote single atoms*. Dissertation, Technische Universität München (2013).
- [55] I. Bongioanni, L. Sansoni, F. Sciarrino, G. Vallone and P. Mataloni. Experimental quantum process tomography of non-trace-preserving maps. *Physical Review A* **82**, 042307 (2010).
- [56] S. Haroche and J.-M. Raimond. *Exploring the Quantum: Atoms, Cavities, and Photons*. Oxford University Press (2013).
- [57] E. Jaynes and F. W. Cummings. Comparison of quantum and semiclassical radiation theories with application to the beam maser. *Proceedings of the IEEE* **51**, 89–109 (1963).

- [58] R. J. Thompson, G. Rempe and H. J. Kimble. Observation of normal-mode splitting for an atom in an optical cavity. *Physical Review Letters* **68**, 1132–1135 (1992).
- [59] L. K. Grover. Quantum mechanics helps in searching for a needle in a haystack. *Physical Review Letters* **79**, 325–328 (1997).
- [60] D. Deutsch and R. Jozsa. Rapid solution of problems by quantum computation. *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences* **439**, 553–558 (1992).
- [61] C. W. Gardiner and M. J. Collett. Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation. *Physical Review A* **31**, 3761–3774 (1985).
- [62] D. F. Walls and G. G. J. Milburn. *Quantum Optics*. Springer (2008).
- [63] H. Kim, R. Bose, T. C. Shen, G. S. Solomon and E. Waks. A quantum logic gate between a solid-state quantum bit and a photon. *Nature Photonics* **7**, 373–377 (2013).
- [64] M. Mücke. *Elektromagnetisch induzierte Transparenz mit einem einzelnen Atom*. Dissertation, Technische Universität München (2011).
- [65] Q. A. Turchette, C. J. Hood, W. Lange, H. Mabuchi and H. J. Kimble. Measurement of conditional phase shifts for quantum logic. *Physical Review Letters* **75**, 4710–4713 (1995).
- [66] Q. Turchette, R. Thompson and H. Kimble. One-dimensional atoms. *Applied Physics B* **60**, (1995).
- [67] E. Freitag and R. Busam. *Funktionentheorie 1. 1*. Springer, Berlin [u.a.] (2000).
- [68] A. Reiserer, C. Nölleke, S. Ritter and G. Rempe. Ground-state cooling of a single atom at the center of an optical cavity. *Physical Review Letters* **110**, 223003 (2013).
- [69] S. Nußmann. *Kühlen und Positionieren eines Atoms in einem optischen Resonator*. Dissertation, Technische Universität München (2006).
- [70] H. J. Metcalf and P. V. d. Straten. *Laser Cooling and Trapping*. Springer (1999).
- [71] R. Grimm, M. Weidemüller and Y. B. Ovchinnikov. Optical dipole traps for neutral atoms. In: B. Bederson and H. Walther, editors, *Advances In Atomic, Molecular, and Optical Physics*, Vol. Volume 42, 95–170. Academic Press, (2000).
- [72] ,D. Steck. Rubidium ^{87}Rb D Line Data. Available at www.steck.us/alkalidata/.
- [73] M. S. Safronova and U. I. Safronova. Critically evaluated theoretical energies, lifetimes, hyperfine constants, and multipole polarizabilities in ^{87}Rb . *Physical Review A* **83**, 052508 (2011).
- [74] B. Arora, M. S. Safronova and C. W. Clark. Magic wavelengths for the np-ns transitions in alkali-metal atoms. *Physical Review A* **76**, 052509 (2007).

- [75] M. S. Safronova, W. R. Johnson and A. Derevianko. Relativistic many-body calculations of energy levels, hyperfine constants, electric-dipole matrix elements, and static polarizabilities for alkali-metal atoms. *Physical Review A* **60**, 4476–4487 (1999).
- [76] C.-Y. Shih and M. S. Chapman. Nondestructive light-shift measurements of single atoms in optical dipole traps. *Physical Review A* **87**, 063408 (2013).
- [77] J. Bochmann, M. Mücke, C. Guhl, S. Ritter, G. Rempe and D. L. Moehring. Lossless state detection of single neutral atoms. *Physical Review Letters* **104**, 203601 (2010).
- [78] I. Dotsenko, W. Alt, S. Kuhr, D. Schrader, M. Müller, Y. Miroshnychenko, V. Gomer, A. Rauschenbeutel and D. Meschede. Application of electro-optically generated light fields for Raman spectroscopy of trapped cesium atoms. *Applied Physics B* **78**, 711–717 (2004).
- [79] L. M. K. Vandersypen and I. L. Chuang. NMR techniques for quantum control and computation. *Reviews of Modern Physics* **76**, 1037–1069 (2005).
- [80] N. Khaneja, T. Reiss, C. Kehlet, T. Schulte-Herbrüggen and S. J. Glaser. Optimal control of coupled spin dynamics: design of NMR pulse sequences by gradient ascent algorithms. *Journal of Magnetic Resonance* **172**, 296–305 (2005).
- [81] G. d. Lange, Z. H. Wang, D. Ristè, V. V. Dobrovitski and R. Hanson. Universal dynamical decoupling of a single solid-state spin from a spin bath. *Science* **330**, 60–63 (2010).
- [82] M. H. Levitt and R. Freeman. NMR population inversion using a composite pulse. *Journal of Magnetic Resonance (1969)* **33**, 473–476 (1979).
- [83] E. D. Black. An introduction to Pound–Drever–Hall laser frequency stabilization. *American Journal of Physics* **69**, 79 (2001).
- [84] D. Bouwmeester, J.-W. Pan, M. Daniell, H. Weinfurter and A. Zeilinger. Observation of three-photon greenberger-horne-zeilinger entanglement. *Physical Review Letters* **82**, 1345–1349 (1999).
- [85] J. Volz, R. Gehr, G. Dubois, J. Estève and J. Reichel. Measurement of the internal state of a single atom without energy exchange. *Nature* **475**, 210–213 (2011).
- [86] C. Schön, E. Solano, F. Verstraete, J. I. Cirac and M. M. Wolf. Sequential generation of entangled multiqubit states. *Physical Review Letters* **95**, 110503 (2005).
- [87] Y. Colombe, T. Steinmetz, G. Dubois, F. Linke, D. Hunger and J. Reichel. Strong atom–field coupling for Bose–Einstein condensates in an optical cavity on a chip. *Nature* **450**, 272–276 (2007).
- [88] D. Hunger, T. Steinmetz, Y. Colombe, C. Deutsch, T. Hänsch and J. Reichel. A fiber Fabry–Perot cavity with high finesse. *New Journal of Physics* **12**, 065038 (2010).

- [89] A. E. Lita, A. J. Miller and S. W. Nam. Counting near-infrared single-photons with 95% efficiency. *Optics Express* **16**, 3032–3040 (2008).
- [90] M. Brune, S. Haroche, V. Lefevre, J. M. Raimond and N. Zagury. Quantum non-demolition measurement of small photon numbers by Rydberg-atom phase-sensitive detection. *Physical Review Letters* **65**, 976–979 (1990).
- [91] C. Guerlin, J. Bernu, S. Deléglise, C. Sayrin, S. Gleyzes, S. Kuhr, M. Brune, J.-M. Raimond and S. Haroche. Progressive field-state collapse and quantum non-demolition photon counting. *Nature* **448**, 889–893 (2007).
- [92] R. Garisto and L. Hardy. Entanglement of projection and a new class of quantum erasers. *Physical Review A* **60**, 827–831 (1999).
- [93] C. F. Roos, M. Riebe, H. Häffner, W. Hänsel, J. Benhelm, G. P. T. Lancaster, C. Becher, F. Schmidt-Kaler and R. Blatt. Control and measurement of three-qubit entangled states. *Science* **304**, 1478–1480 (2004).
- [94] J. Cho and H.-W. Lee. Generation of atomic cluster states through the cavity input-output process. *Physical Review Letters* **95**, 160501 (2005).
- [95] V. Giovannetti, S. Lloyd and L. Maccone. Quantum-enhanced measurements: beating the standard quantum limit. *Science* **306**, 1330–1336 (2004).
- [96] C. Simon and W. T. M. Irvine. Robust long-distance entanglement and a loophole-free bell test with Ions and photons. *Physical Review Letters* **91**, 110405 (2003).
- [97] J. Hofmann, M. Krug, N. Ortegel, L. Gérard, M. Weber, W. Rosenfeld and H. Weinfurter. Heralded entanglement between widely separated atoms. *Science* **337**, 72–75 (2012).
- [98] D. L. Moehring, P. Maunz, S. Olmschenk, K. C. Younge, D. N. Matsukevich, L.-M. Duan and C. Monroe. Entanglement of single-atom quantum bits at a distance. *Nature* **449**, 68–71 (2007).
- [99] H. Bernien, B. Hensen, W. Pfaff, G. Koolstra, M. S. Blok, L. Robledo, T. H. Taminiau, M. Markham, D. J. Twitchen, L. Childress and R. Hanson. Heralded entanglement between solid-state qubits separated by three metres. *Nature* **497**, 86–90 (2013).
- [100] M. Paris and J. Řeháček. *Quantum state estimation*. Springer (2004).
- [101] J. L. O’Brien, G. J. Pryde, A. Gilchrist, D. F. V. James, N. K. Langford, T. C. Ralph and A. G. White. Quantum Process Tomography of a Controlled-NOT Gate. *Physical Review Letters* **93**, 080502 (2004).
- [102] G. S. Fishman. *Monte Carlo concepts, algorithms, and applications*. Springer New York, New York, NY (1996).

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